

Week 02: Grids & Layouts

COMS 4170 • Spring 2026

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Tuesday January 27, 2026



COMS 4170 · SPRING 2026

Review



10 Usability Heuristics

1 Visibility of System Status

Designs should **keep users informed** about what is going on, through appropriate, timely feedback.

2 Match between System and the Real World

The design should speak the users' language. Use words, phrases, and concepts **familiar to the user**, rather than internal jargon.

3 User Control and Freedom

Users often perform actions by mistake. They **need a clearly marked "emergency exit"** to leave the unwanted state.

4 Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. **Follow platform conventions.**

5 Error Prevention

Good error messages are important, but the best designs **prevent problems** from occurring in the first place.

6 Recognition Rather Than Recall

Minimize the user's memory load by making elements, actions, and options visible. Avoid making users remember information.

7 Flexibility and Efficiency of Use

Shortcuts — hidden from novice users — may **speed up the interaction** for the expert user.

8 Aesthetic and Minimalist Design

Interfaces should not contain information which is irrelevant. Every extra unit of information in an interface **competes** with the relevant units of information.

9 Recognize, Diagnose, and Recover from Errors

Error messages should be expressed in **plain language** (no error codes), precisely indicate the problem, and constructively suggest a solution.

10 Help and Documentation

It's best if the design **doesn't need** any additional explanation. However, it may be necessary to provide documentation to help users understand how to complete their tasks.

The main feature of many websites is to **display information** in a way that helps users **accomplish a goal**.

```
i:Exit --:PrevPg <Space>NextPg v:View Attachm. d:Del r:Reply j:Next ? :Help
624 Aug 03 T Martinez ( 37) Loans with tiny points are here now
625 0 Jul 01 R. Jackson ( 123) Loans with tiny rates are here now
626 Aug 05 Benjamin E. Mag ( 50) Long time no hear
627 May 17 Krista Aaron ( 44) long time no see....
628 0 Jun 03 Josiah House ( 35) Looking for a hot date tonight, tomorrow, or next week?
629 Jul 03 Brigitte I. Hay ( 63) Looking for a N.ew H.ome?
630 May 17 Joe Burns ( 58) Looking for you
631 Jun 01 Save in a poor ( 145) Low Rate Consolidation Mortgage Loan
632 + Jul 02 Igiel@virtualig ( 2) LowCost SoftWare OnCD
-- Mutt: Mail/junk/spam [Msgs:950 Old:142 10M]-- (subject/date)----- (66%)
Date: Mon, 17 May 2004 03:40:09 +0100
From: Krista Aaron <Christinefeminine@highstream.com>
Subject: long time no see....

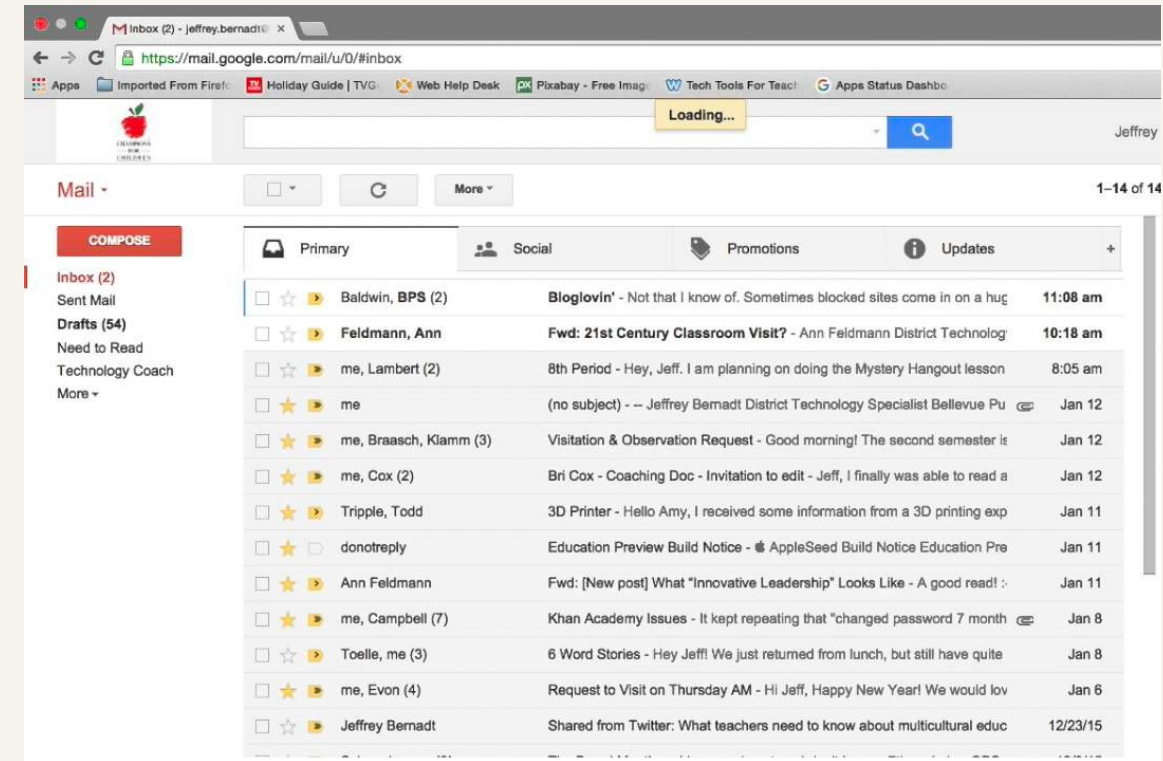
[-- Autoview using /usr/bin/links -force-html -dump '/tmp/mutt.html' --]
My name is Jen and I'm new to this dating thing. I've checked out your profile
you put up and it's interesting. => I just want to get to know you a little
better if you don't mind, come check my profile out at:

www.livejen.com/chat.html

I also got a webcam so we can make it interesting, anyways hope you get back to
me.
bye :)

gxsnkxxgnduvyjjwceudc.jobxs
zcocccrociessbehgbpow
rnxlfu.jnqpbllipdkgwuyqofracsz
xmqaubxsbjrppeibvlpfhqowldtp
bixhghvrxrtggfeocofzyzb
hugzffaffulsklpzhrfxbt
btpztlfotqmmoaiwlosqv
-- 627/950: Krista Aaron long time no see.... -- (69%)
Key is not bound. Press '?' for help.
```

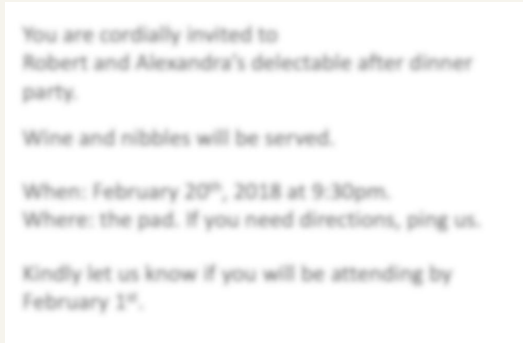
Pine text-based email client, 1989



GMail free email service, 2004



Seven tools for visually indicating importance



Conceptual grouping



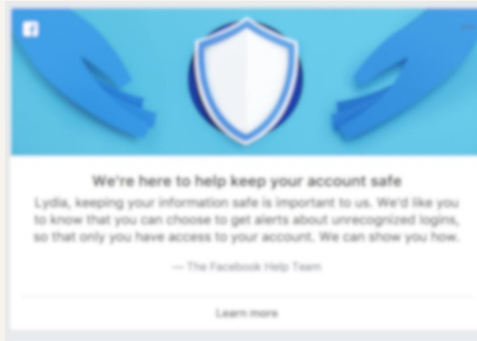
Location



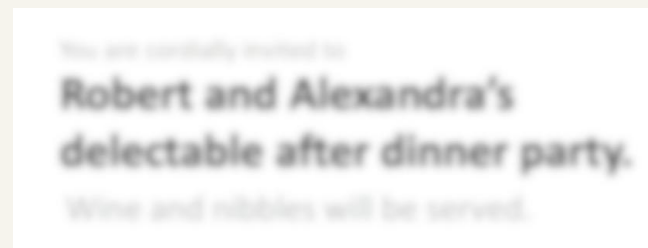
Whitespace



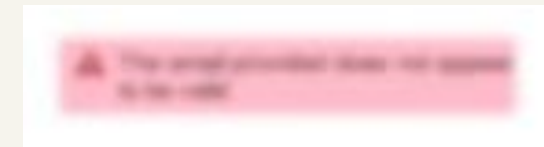
Size



Images



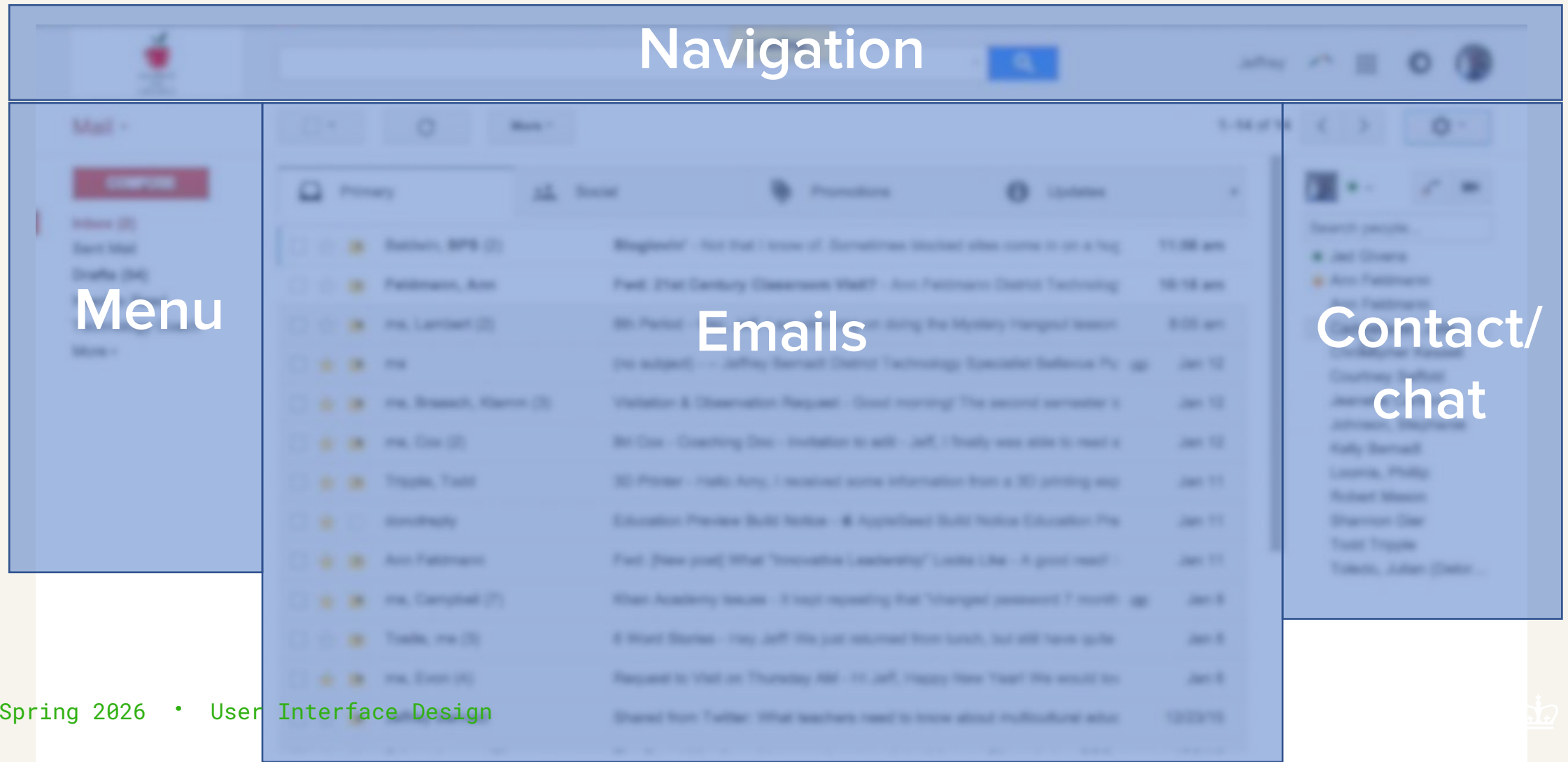
Contrast



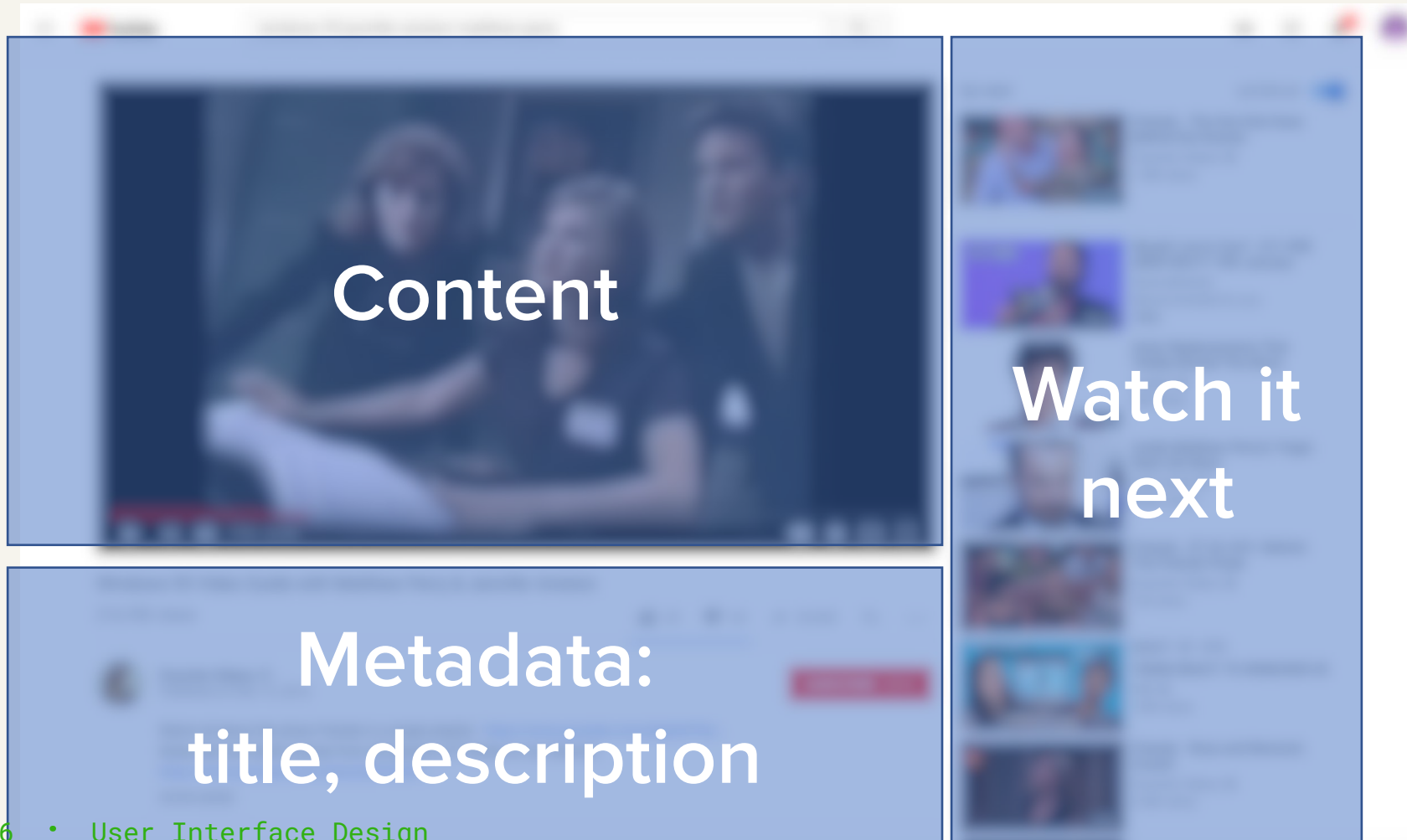
Color



Conceptual groups in Gmail



Conceptual groups in YouTube



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How should we layout information on a page?





A L A C A R T E

APPETIZERS

Home Made Italian Antipasto	35
Italian Ham	25
Imported Fillet of Anchovies	25
Italian Salami	25
Mortatella di Bologna	20
Sea Food Cocktail (in season)	20
Celery and Olives	20
Italian or French Sardines	30
Italian Antipasto (in cans)	40
Tuna Fish Salad	35
Artichoke Salad with Olive Oil	35

SALADS

Lettuce and Tomato Salad	20
Combination Salad	25
Lobster Salad	35
Hearts of Lettuce	15
Avocado Salad	25
Shrimp Salad	35
Orange Salad with Whipped Cream	25
Tomato Salad	15
Chicken Salad	35

STEAKS AND CHOPS

Broiled Tenderloin Steak	
Mushroom Sauce	90
Sirloin Steak—	
Bordeleaise Sauce	75
T-Bone Steak	75
French Lamb Chops	50
Veal or Pork Chops	45

POULTRY

½ Spring Chicken—Broiled	60
Chicken Cacciatore	70
Chicken Sauté—Mushrooms	75
½ Fried Spring Chicken	60

Bread, Butter, Potatoes, Fresh Vegetables
Served With Above Orders.

EGGS AND OMELETS

Jelly Omelet	40
Salami Omelet	40
Cheese Omelet	40
Onion Omelet	40
Spinach Omelet	40
Ham Omelet	40
Bacon and Eggs	40
Ham and Eggs	40
Pork Sausage and Eggs	40
Poached Eggs on Toast	30
Poached Eggs Viennese	35
Fried Ham per Order	35

Bread, Butter or Toast and Potatoes
With Above Orders

COLD SANDWICHES

Tongue Sandwich	15
American Cheese Sandwich	15
Roast Beef Sandwich	20
Italian Ham and Cheese	20
Lettuce and Tomato Sandwich	15
Ham and Tomato Sandwich	20
Sliced Cold Chicken Sandwich	40
Cold Pork Sandwich	25
Corned Beef Sandwich	15

SEA FOOD IN SEASON

Lobster Louis	45
Broiled Lobster	45
Lobster Newburg	60
Crab Meat	50
Oysters Stew	35
Fried Oysters	45

CHEESE

Roquefort Cheese	25
Provalotta Cheese	25
Local Swiss	15
Bel Paese	25

DRINKS

Coffee	05
Hot Tea	05
Milk	25
Buttermilk	05
Eastern Beer, pint	20
Local Beer, pint	15

DESSERTS

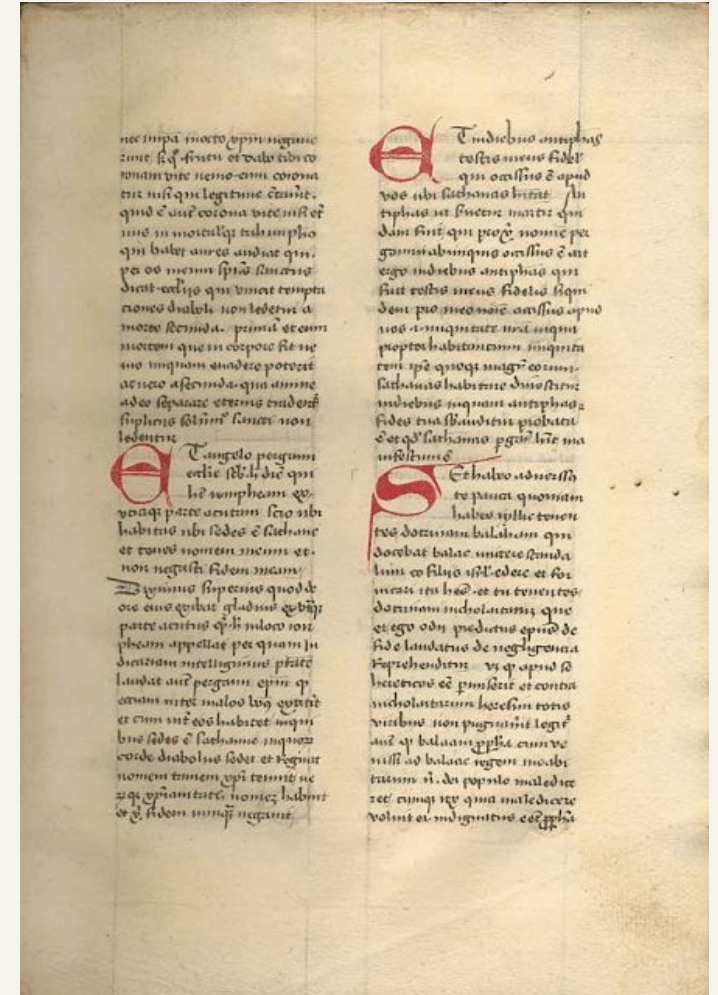
Pie, Ice Cream, Sherbet, Jello, Custard or Pudding	
---	--

HOT SANDWICHES

Club House Sandwich	40
Hamburger Sandwich	15
Ham and Egg Sandwich	25
Bacon and Egg Sandwich	25
Denver Sandwich	25
Hot Roast Beef	25
Melted Cheese Sandwich	20
Fried Ham with Tomato	20
Bacon and Tomato	25
Sirloin Steak Sandwich	35

Eat and drink well and you will live long!

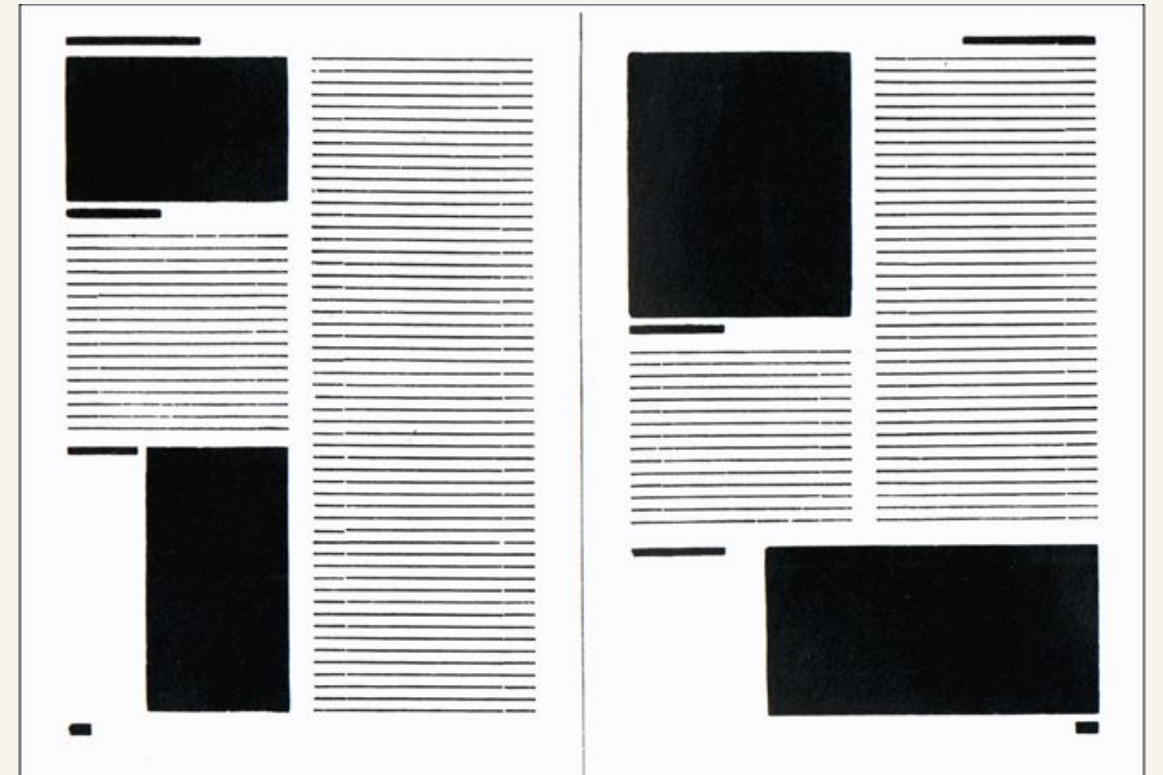
Symmetrical



Then: Symmetric Layout



Now: Asymmetric Layout



The Bauhaus School of Design (1919–1933)



Modern Grid

German typographer Jan Tschichold made grid layouts inspired by the Bauhaus

	Lorem Ipsum!	Lorem Ipsum!	Lorem Ipsum!	
	<i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	<i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	<i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	

	Lorem Ipsum Dolor Sit Amet Consectetur	Lorem ipsum dolor sit amet Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	
		Lorem ipsum dolor sit amet Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	
		Lorem ipsum dolor sit amet Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	



Grids can convey importance through...

grouping, location, size and whitespace

You are cordially invited to
Robert and Alexandra's delectable after dinner
party.

Wine and nibbles will be served.

When: February 20th, 2018 at 9:30pm.
Where: the pad. If you need directions, ping us.

Kindly let us know if you will be attending by
February 1st.

**Conceptual
grouping**



Location



Whitespace



Size



Grids can convey importance through...

grouping, location, size and whitespace

	Lorem Ipsum! <i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	Lorem Ipsum! <i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	Lorem Ipsum! <i>dolor sit amet</i> Consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	
--	--	--	--	--

How do we create groups in a grid?

By creating a column or row.

How do we use location to make a group appear more important?

Put the information in the upper left grid cell.

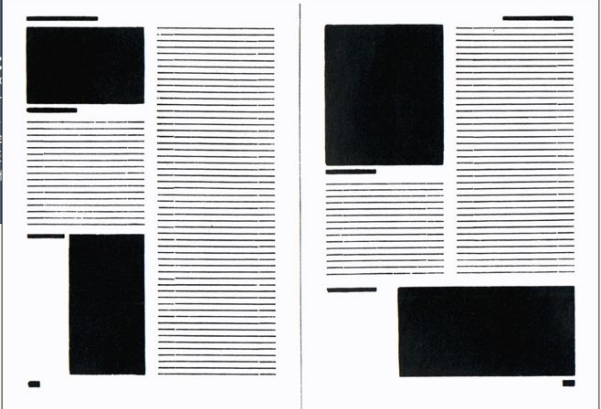
How do we use size to make a group appear more important?

Make a column wider.

How do we use whitespace to make a group appear more important?

Increase the size of the gutters.

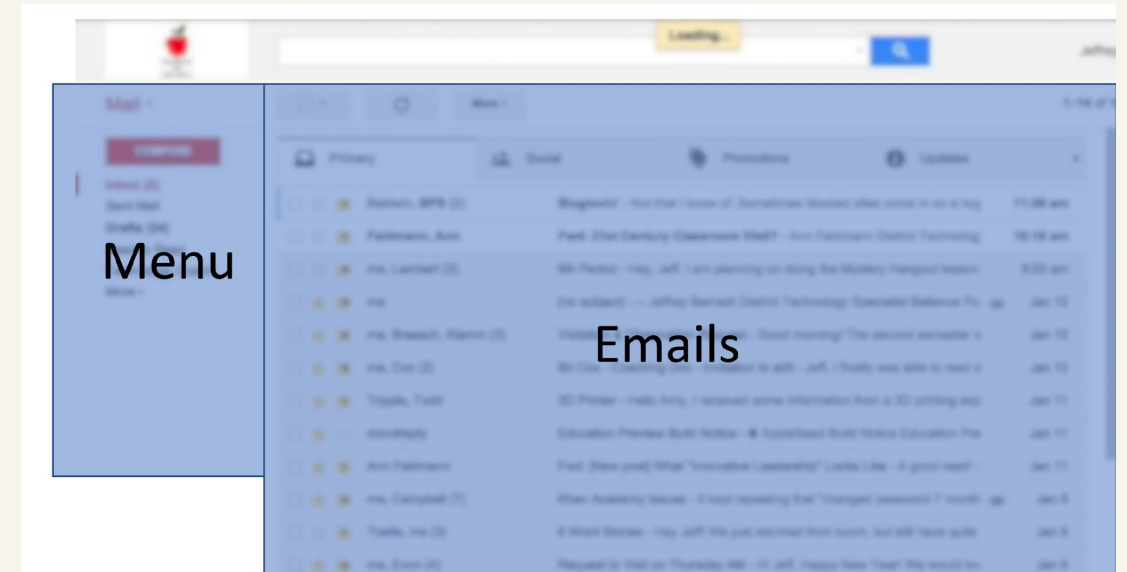
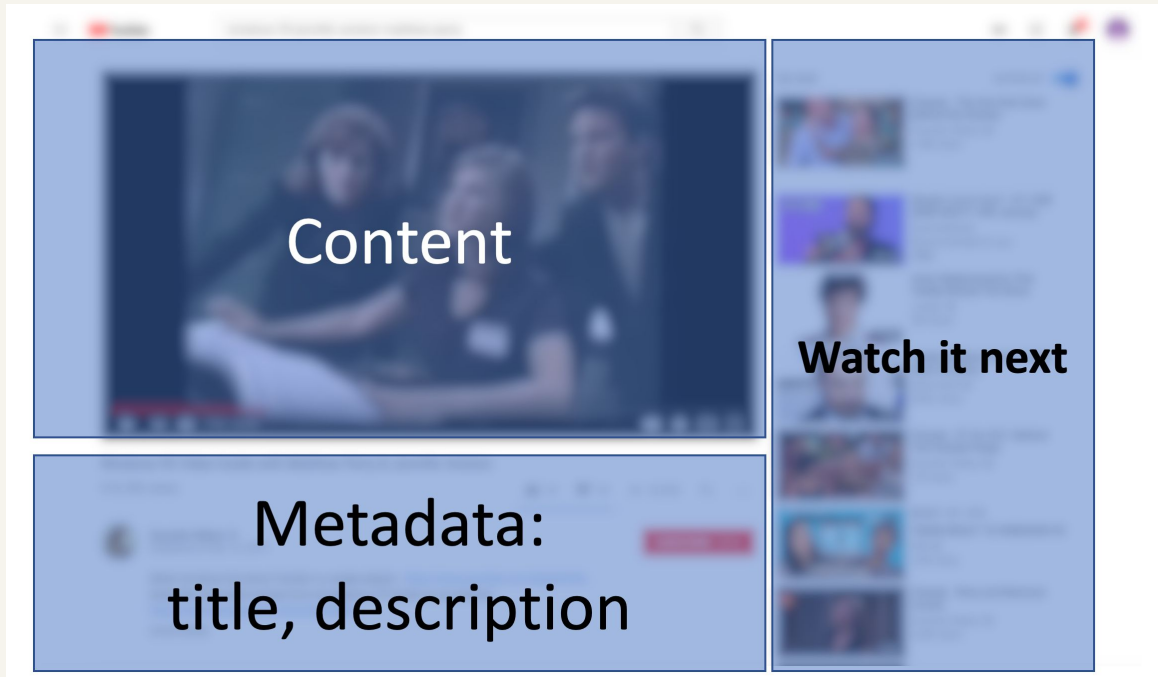
Not all the ideas of the Bauhaus school are new



But there is freedom from the symmetric layout.



Grid layout heavily influenced web layout



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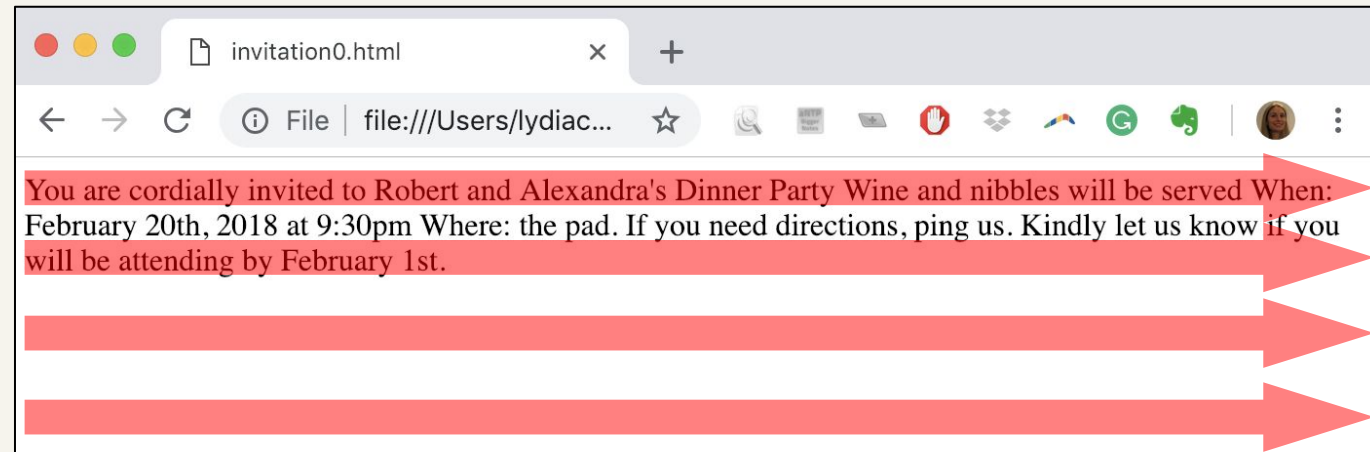
Basic Layout in HTML (no grids yet)

How to construct the layout of information in
HTML and CSS



Web browsers are in charge of rendering HTML

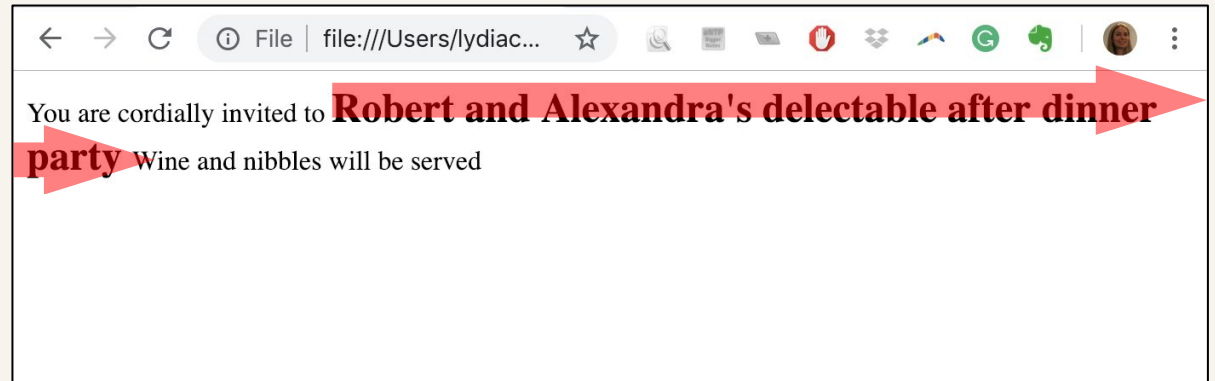
```
invitation0.html
1 <html>
2   <body>
3
4     You are cordially invited to
5     Robert and Alexandra's Dinner Party
6     Wine and nibbles will be served
7
8     When: February 20th, 2019 at 9:30pm
9     Where: the pad. If you need directions, ping us.
10
11     Kindly let us know if you will be attending by February 1st.
12
13
14   </body>
15 </html>
```



When you add HTML elements, they either obey the line-wrap layout or break it.

```
19 <div>
20   You are cordially invited to
21   <span class="dark big">
22     Robert and Alexandra's
23     delectable after dinner party
24   </span>
25   Wine and nibbles will be served
26 </div>
```

**** elements **obey**
line-wrap layout



```
<div>
  You are cordially invited to <span class="dark big">
</div>
<div>
  When: February 20th, 2019 at 9:30pm Where: the pad. If you need
</div>
```

You are cordially invited to **Robert and Alexandra's**

When: February 20th, 2019 at 9:30pm Where: the pad. If you need

<div> elements **break** line-wrap layout



When you use a new element, ask: “Does it obey or break line-wrap layout?”

Anchor
(hyperlink)

```
<a href="slides.pdf">Slides are here</a>
```

[Slides are here](#)

Image

```

```



The best way to know what a layout will look like is to run it.

Form

```
<form>  
  <input type="text" name="firstname" value="Mickey">  
  <br>  
  Last name:<br>  
  <input type="text" name="lastname" value="Mouse">  
  <br><br>  
  <input type="submit" value="Submit">  
</form>
```

First name:
Mickey
Last name:
Mouse
Submit

Today's goal:

Remake this information layout using HTML and CSS.

facebook



PictureCorrect Photography Tips

What is your Photography-Related New Year's Resolution??

15 minutes ago · Like · Comment

👍 Adel Gomaa and Richard Schneider like this.



Sadi Molloy get my business looking more professional: car signed: tick, making up portrait folders: tick, stickers & cd covers: still to do... its getting there though.

11 minutes ago · Flag



PictureCorrect Photography Tips My resolution for 2011 is to capture more landscape photos outside the U.S. and add more countries to my portfolio! What is yours?

-Richard

11 minutes ago · Flag



James Donnelly Larger concerts to cover and improve my portrait editing style/skill

10 minutes ago · Flag



Jose Antonio Rubido To push my photpography foward full throttle

9 minutes ago · Flag



Deacon Jonathan Jerome Johnson To learn more and enhance the quality of my shots!

5 minutes ago · Flag



Latu Peti Vimahi Manulua-hafoka learn learn learn!!!

4 minutes ago · Flag

“I finished coding up my website...”

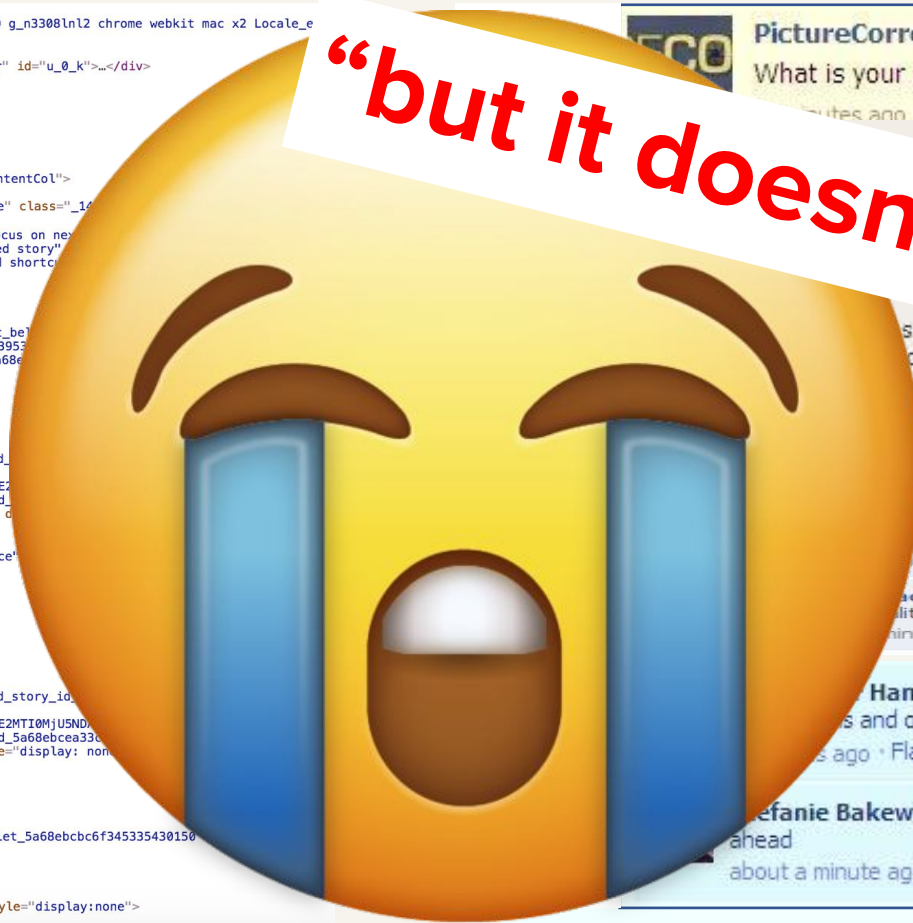
```

<!--></head></>
<body class="hasLeftCol_2yq home composerExpanded_5vb_fbx s_2ltu_605a_lfs_61s0 g_n3308ln1p chrome webkit mac h2 Locale_e
cores-gte4 hasANavMenuabar" dir="ltr" data-gr-c-s-loaded="true">
<div class="l1" id="u_0_1">
<div class="3_0 1toe 3_1 3_1 uiBoxGray noborder" data-testid="ax-navigation-menuabar" id="u_0_k"></div>
<div id="pagelet_bluebar" data-referrer="pagelet_bluebar"></div>
<div id="globalContainer" class="uiContextualLayerParent">
<div class="fb_content clearfix" id="content" role">
<div class="2pie">
<div id="toolbarContainer"></div>
<div class="clearfix" id="mainContainer">
<div class="lqkq_lqks" id="leftCol" style="margin-top: -12px;"></div>
<div class="5r homeviewerContent homeFixedLayout newsFeedComposer_lqkq_lqlq" id="contentCol">
<div class="clearfix" id="content_container">
<div id="rightCol" role="complementary" aria-label="Reminders, trending topics and more" class="1415">
<div id="contentArea" role="main" class="1415_lqkq_lqkx">
<div id="stream_pagelet" data-referrer="stream_pagelet" data-at-shortcutkeys="["j":"focus on new
story","p":"post a new status","l":"like/unlike selected story","c":"comment on selected story"
selected story's attachment","/":"search","q":"search chat contacts","?":"show keyboard shortcuts"]">
<div class="2c44"></div>
<div id="pagelet_megaphone" data-referrer="pagelet_megaphone"></div>
<div id="pagelet_composer" data-referrers="pagelet_composer"></div>
<div id="pagelet_prompts" data-referrers="pagelet_prompts"></div>
<div class="pagelet" id="stories_pagelet_below_composer" data-referrers="stories_pagelet_b
<div id="topnews_main_stream_408239535924329" data-referrer="topnews_main_stream_40823953
<div class="5pcb_dp7_4j3f" aria-label="News Feed" role="region" id="feed_stream_5a68
<h2 class="accessible_elem" id="newsFeedHeading">News Feed</h2>
<div role="feed">
<div class="41kz" id="substream_0" data-referrer="substream_0">
<div>
<div id="u_ps_0_0">
<div>
<div data-fte="1" data-ftr="1" class="5jmm_5pat_3lb4_s_n3308ln1p" id="hyperfeed_story_id"
testid="fbfeed_story" data-cursor=
"MTUxNjg5NTU0NzoxNTE2ODIINTQ3QjU6LTlTY0NTk0OTY0Tk0TY0NTgWjIGMTUxNTE4MDA4YjoxNTE2ODI0EzE
-3733431725408607942" data-timestamp="1516764239" data-referrer="hyperfeed_story_id"
aria-setsize="8" role="article" aria-labelledby="js_o" aria-describedby="js_p_js_q" d
<div class="4_u2 mbm_4mrt_5v3q_4_u8" id="u_ps_0_0_1">
::before
<div class="3ccb" data-ft="{"tn":"R"}" data-gt="{"type":"click2canvas","fbsource"
</div></div>
</div>
</div>
</div>
</div>
</div>
<div data-fte="1" data-ftr="1" class="5jmm_5pat_3lb4_s_n3308ln1p" id="hyperfeed_story_id"
testid="fbfeed_story" data-cursor=
"MTUxNjg5NTU0NzoxNTE2ODIINTQ3QjU6LTlTY0NTk0OTY0Tk0TY0NTgWjIGMTUxNTE4MDA4YjoxNTE2ODI0EzE
-645949299086858062" data-timestamp="1516677883" data-referrer="hyperfeed_story_id_5a68ebcea330
aria-setsize="8" role="article" aria-labelledby="js_r" aria-describedby="js_s" style="display: none"
</div>
</div>
</div>
</div>
<div data-fte="1" data-ftr="1" class="5jmm_5pat_3lb4_s_n3308ln1p" id="hyperfeed_story_id"
testid="fbfeed_story" data-cursor=
"MTUxNjg5NTU0NzoxNTE2ODIINTQ3QjU6LTlTY0NTk0OTY0Tk0TY0NTgWjIGMTUxNTE4MDA4YjoxNTE2ODI0EzE
-645949299086858062" data-timestamp="1516677883" data-referrer="hyperfeed_story_id_5a68ebcea330
aria-setsize="8" role="article" aria-labelledby="js_r" aria-describedby="js_s" style="display: none"
</div>
</div>
</div>
</div>
<div class="41kz" id="substream_1" data-referrers="substream_1"></div>
<div id="more_pager_pagelet_5a68ebcbcf345335430150" data-referrer="more_pager_pagelet_5a68ebcbcf345335430150"
</div>
</div>
</div>
</div>

</div>
::after

```

“but it doesn’t work yet!”



How would you help this person?



Iterative Style of Programming

What is the **smallest** unit of progress I can make?

Iteration

```
<html>
  <body>
    Hello world!
  </body>
</html>
```

Hello World!

Goal



Does it look ok?



Step 0. Make an HTML page & connect the CSS

```
<!DOCTYPE html>
<html>
<head>
<style>
  #test{
    color: blue;
  }
</style>
</head>

<body>
  <div id="test">Hello World!</div>
</body>
|
</html>
```

Hello World!



What should we do first?



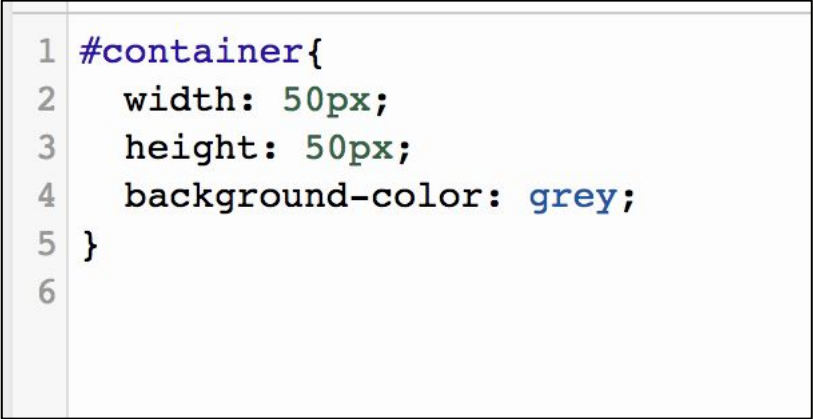
**Make a
container**



Step 1: Make a container <div>



```
<div id="container">  
  
</div>
```



```
1 #container{  
2   width: 50px;  
3   height: 50px;  
4   background-color: grey;  
5 }  
6
```

Iterative Style of Programming

What's the most **basic** thing I can do to make **necessary** progress towards my goal?

Iteration

```
<html>
  <body>
    Hello world!
  </body>
</html>
```

Hello World!

Goal

facebook



Does it look ok?



Does it look ok? What's next?



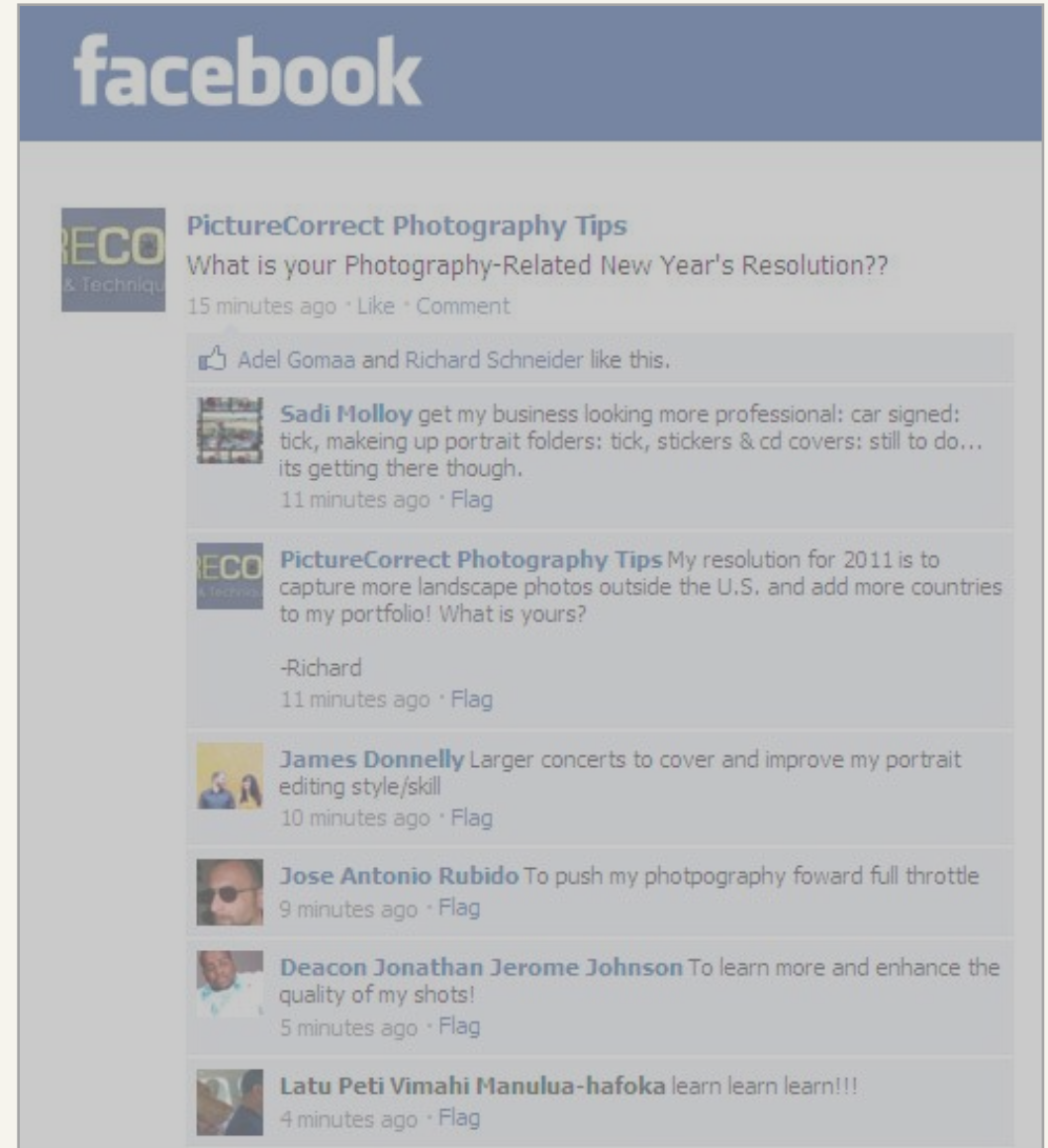
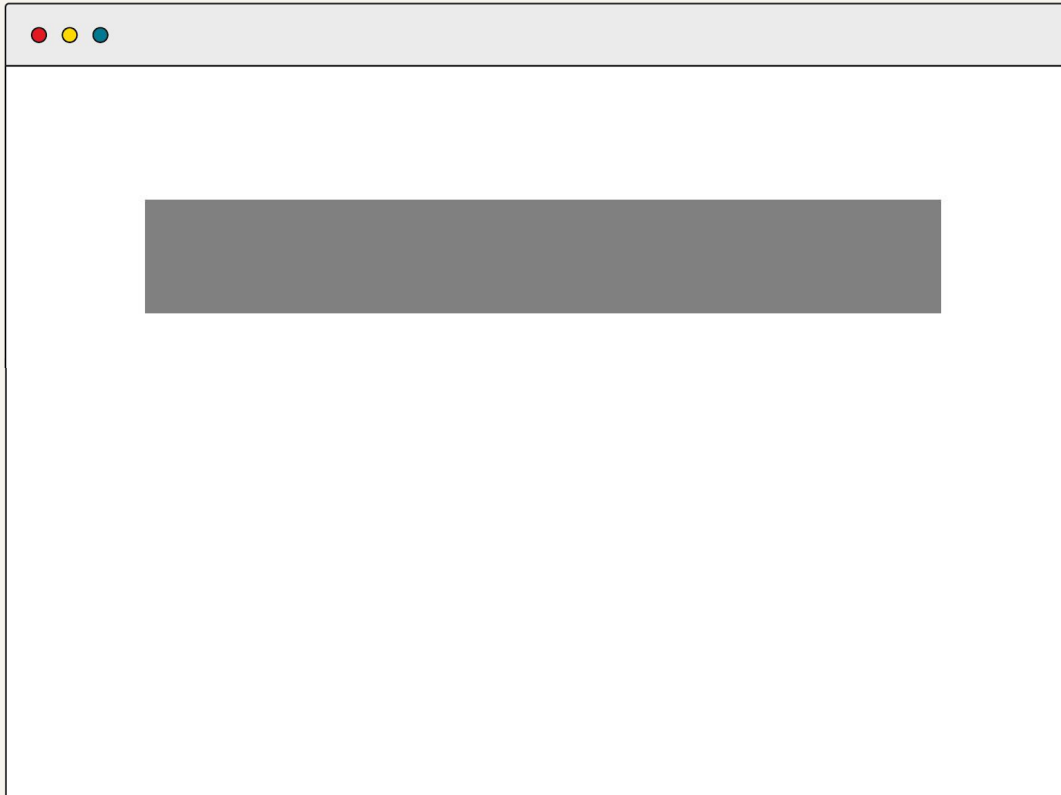
Give the container `<div>` a realistic width.

```
<div id="container">  
  
</div>
```

```
#container {  
  width: 350px;  
  height: 50px;  
  margin: 50px;  
  background-color: grey;  
}
```



What's next?

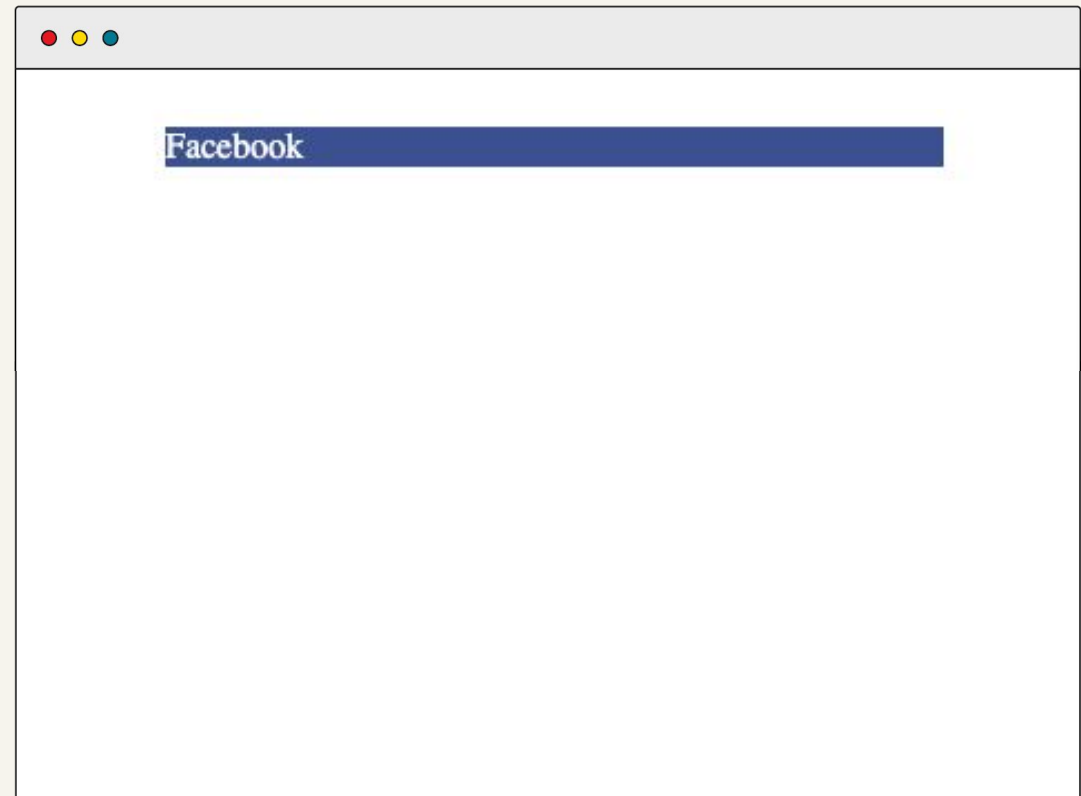


Create the blue Facebook header

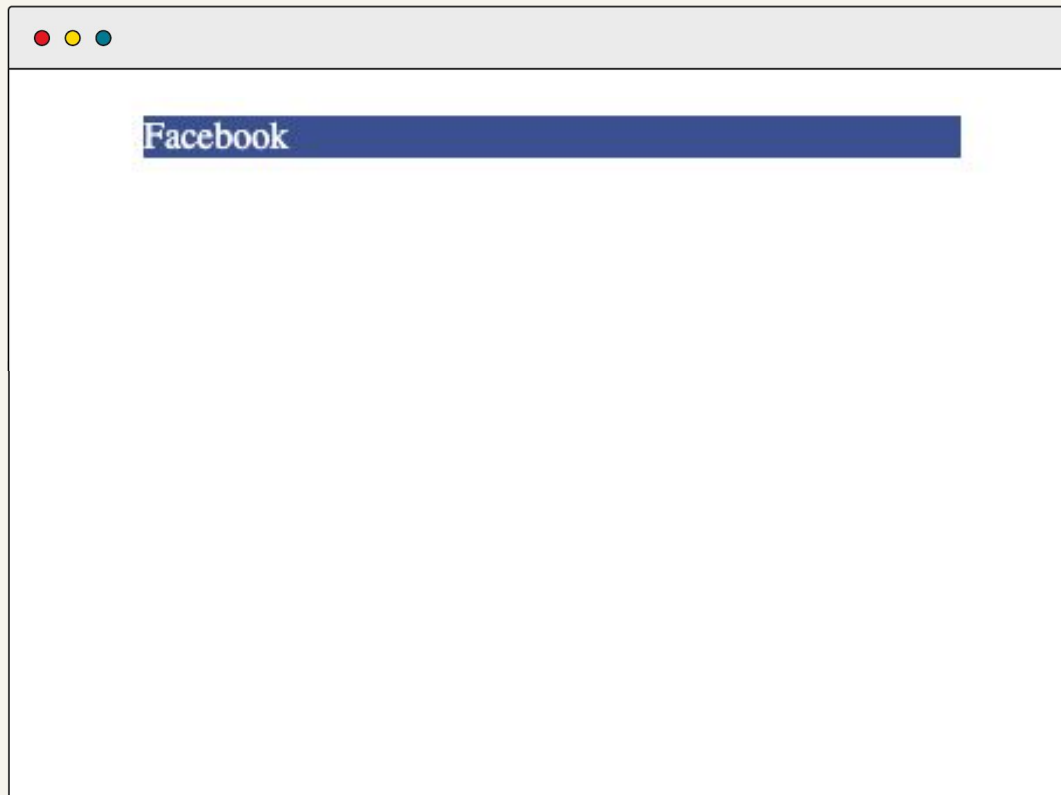
```
<div id="container">
  <div id="fb_brand">
    Facebook
  </div>
</div>
```

```
#container {
  width: 350px;
  height: 50px;
  margin: 0 auto;
}

#fb_brand {
  background-color: rgb(67,90,156);
  color: white;
}
```



What next?

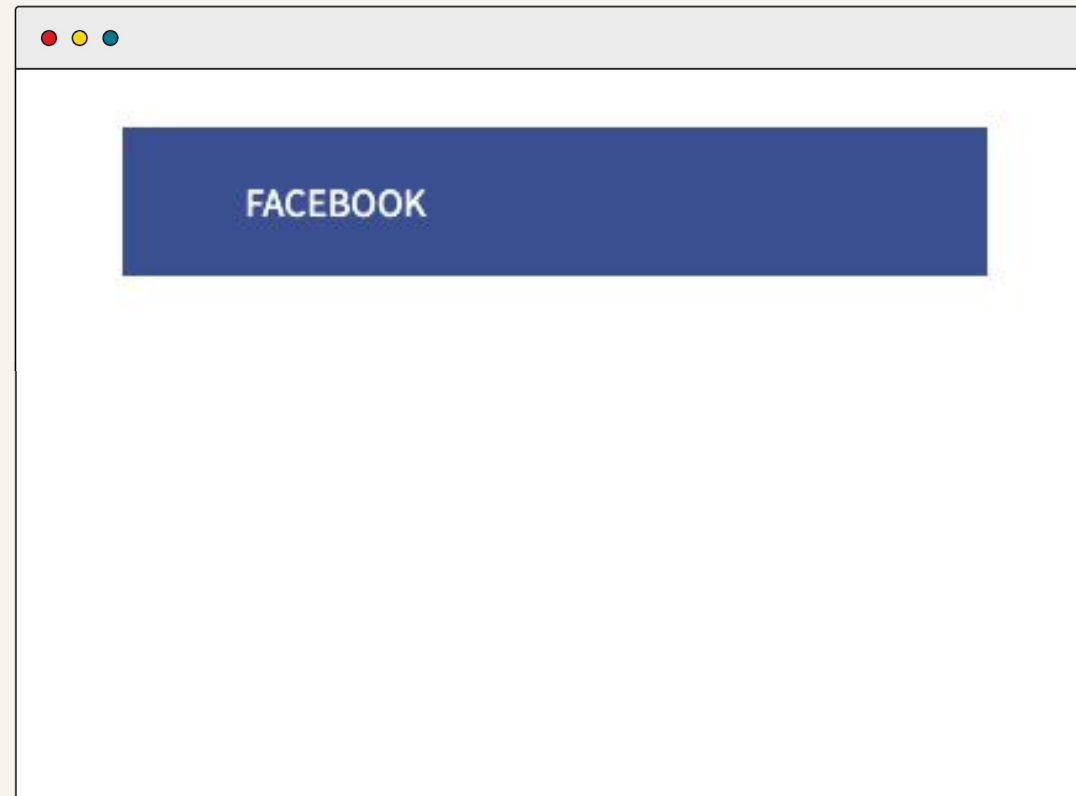


Get the Header height right. Add padding.

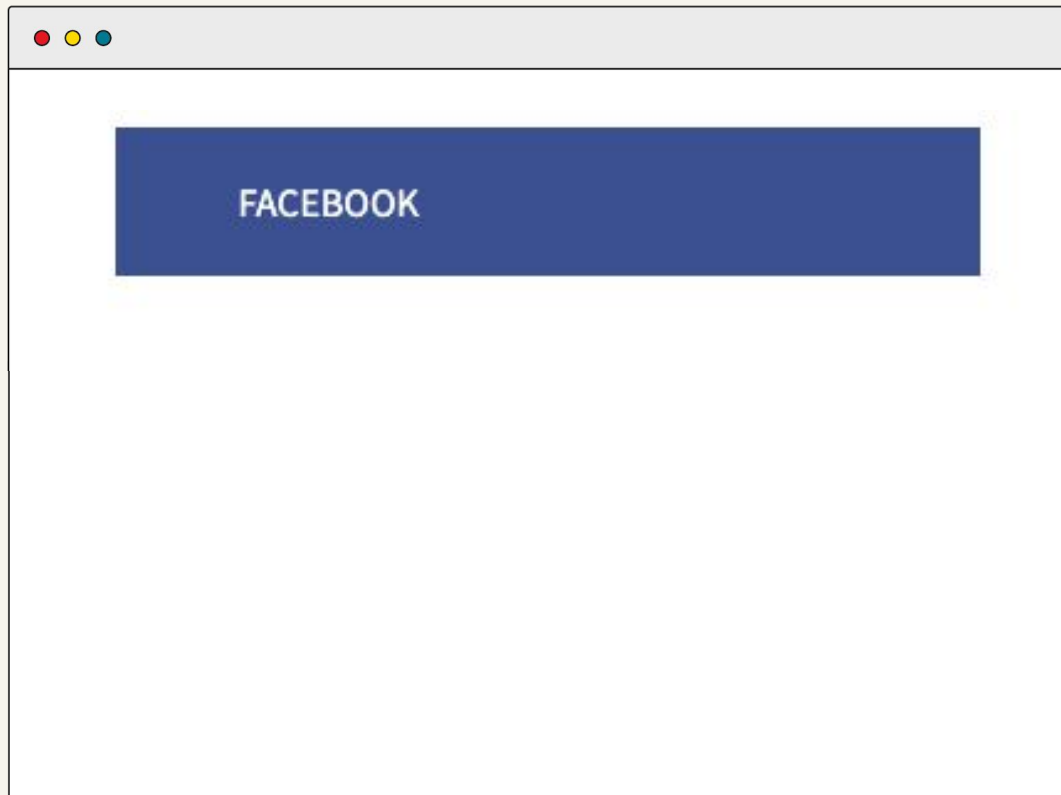
```
<div id="container">
  <div id="fb_brand">
    Facebook
  </div>
</div>
```

```
#container {
  width: 350px;
  height: 50px;
  font-family: 'Source Sans Pro', sans-serif;
  margin: 0 auto;
}

#fb_brand {
  background-color: rgb(67,90,156);
  color: white;
  padding-left: 50px;
  padding-top: 20px;
  padding-bottom: 20px;
  text-transform: uppercase;
}
```

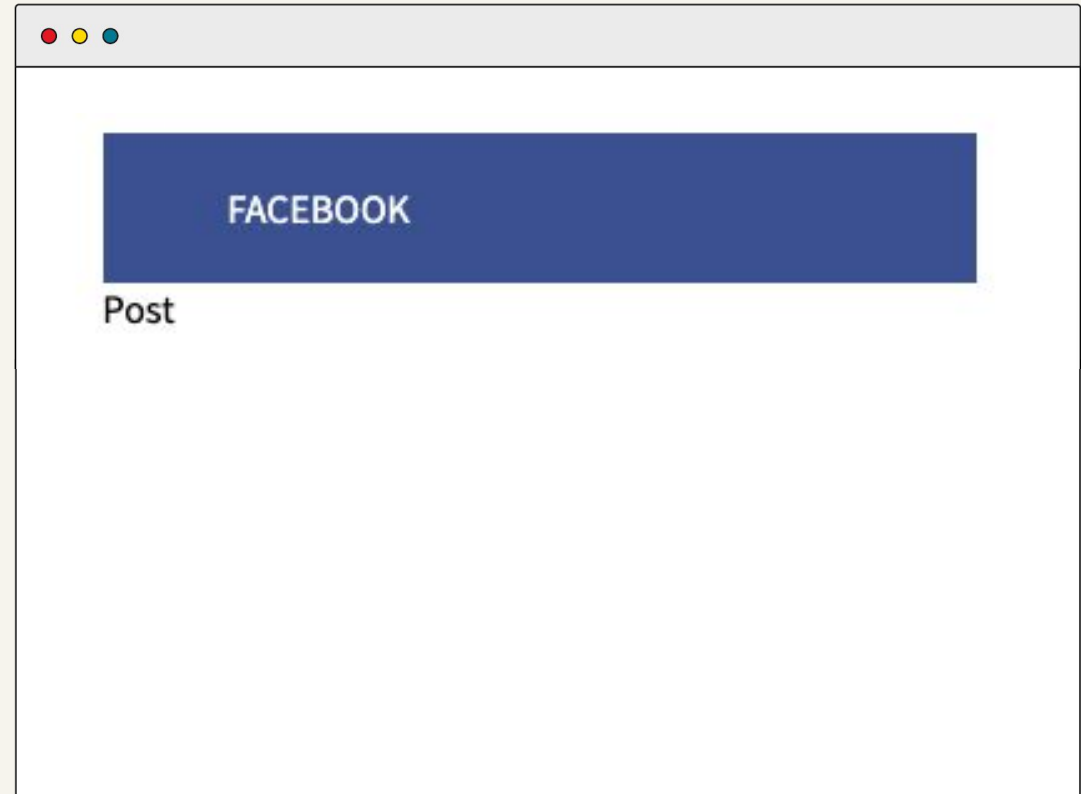


What next?

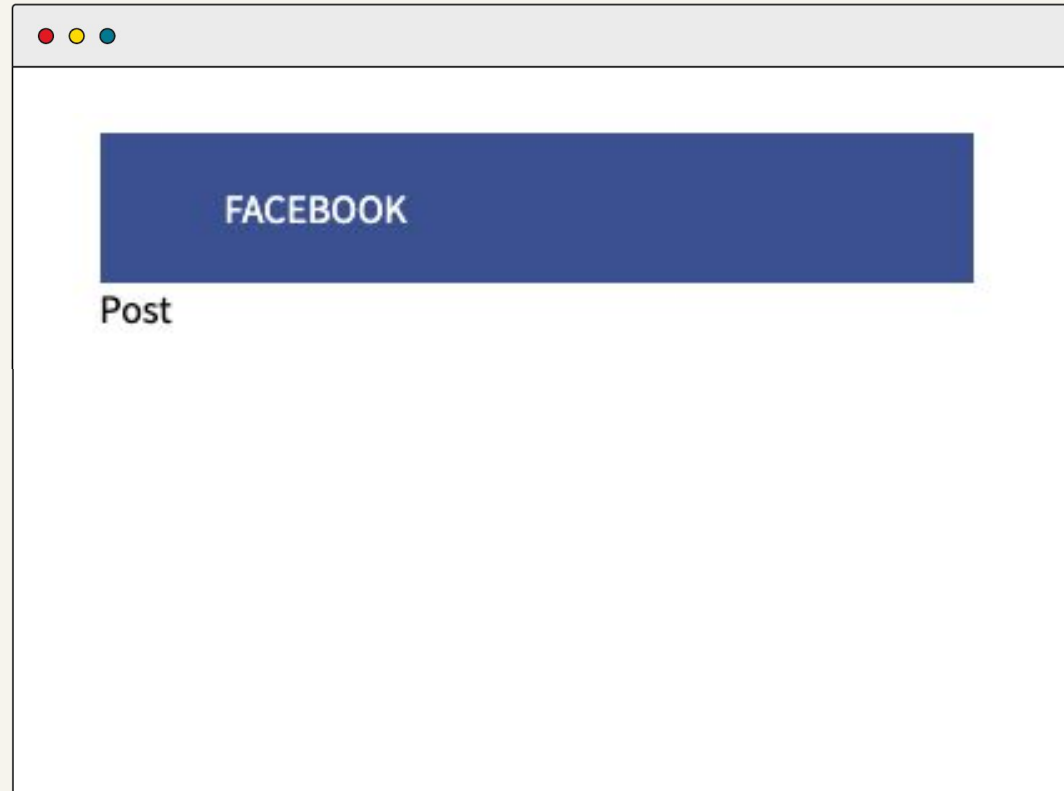


Add the first post

```
<div id="container">  
  <div id="fb_brand">  
    Facebook  
  </div>  
  <div id="top_post">  
    Post  
  </div>  
</div>
```



What next?



Add internal structure to the first post

```
<div id="top_post">
  <div id="top_post_image"></div>
  <div id="top_post_text">
    Top Post Text
  </div>
</div>
```

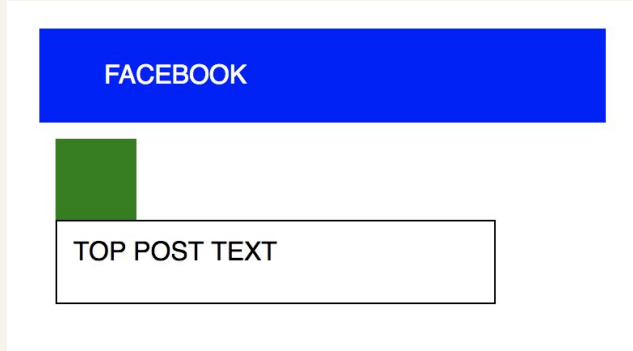
```
#top_post_image {
  height: 50px;
  width: 50px;
  background-color: green;
  display: inline-block;
}

#top_post_text {
  height: 50px;
  width: calc(100% - 50px);
  background-color: lightcoral;
  display: inline-block;
  vertical-align: top;
  line-height: 50px;
}
```



By default, `<div>` elements flow down the page, via **block**

To get them to flow across the page, use **inline-block**



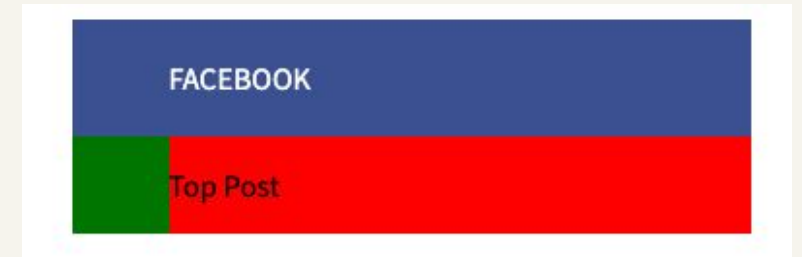
```
18 #top_post_image{
19   height: 50px;
20   width: 50px;
21   background-color: green;
22   /*display: inline-block;*/
23 }
24
25 #top_post_text{
26   height: 30px;
27   width: 250px;
28   padding: 10px;
29   border: 1px solid black;
30   /*display: inline-block;*/
31   vertical-align: top;
32 }
```

Normal flow is the default behavior of the web browser. Block level elements stack on top of each other.

```
div {
  display: block;
}
```

vs

```
div {
  display: inline-block;
}
```



```
#top_post_image {
  height: 50px;
  width: 50px;
  background-color: green;
  display: inline-block;
}

#top_post_text {
  height: 50px;
  width: calc(100% - 50px);
  background-color: lightcoral;
  display: inline-block;
  vertical-align: top;
  line-height: 50px;
}
```



Normal Flow

Normal flow is where you begin with any layout. In the normal flow, text elements are laid out from top to bottom, and from left to right in left-to-right reading languages (or from right to left in right-to-left reading languages). This is the default behavior of the web browser.

In the normal flow, block-level elements stack on top of one another and inline elements fill the available space.



display

display controls the behavior of the box in which content sits. We'll cover several of the most commonly used values of the **display** property:

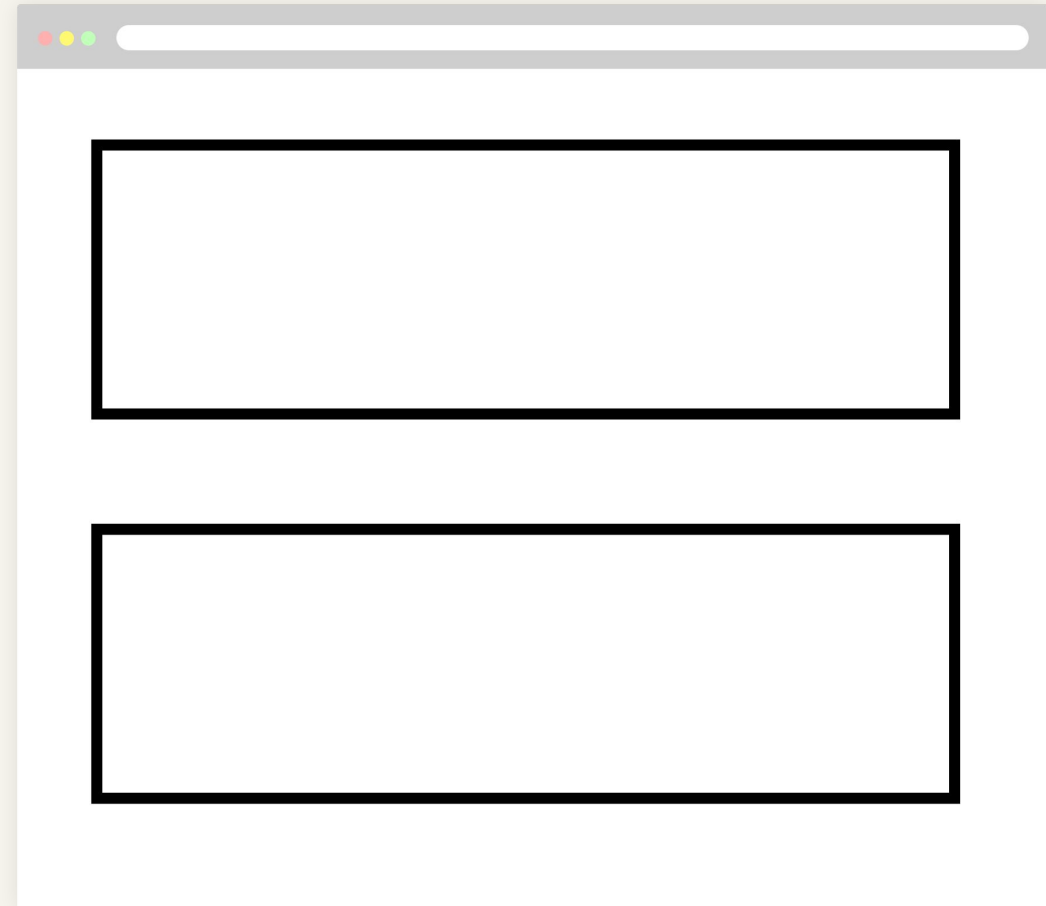
- **block**
- **inline**
- **inline-block**



display: block

This element takes up as much width as possible and the following element drops to a new line.

```
div {  
    display: block;  
}
```

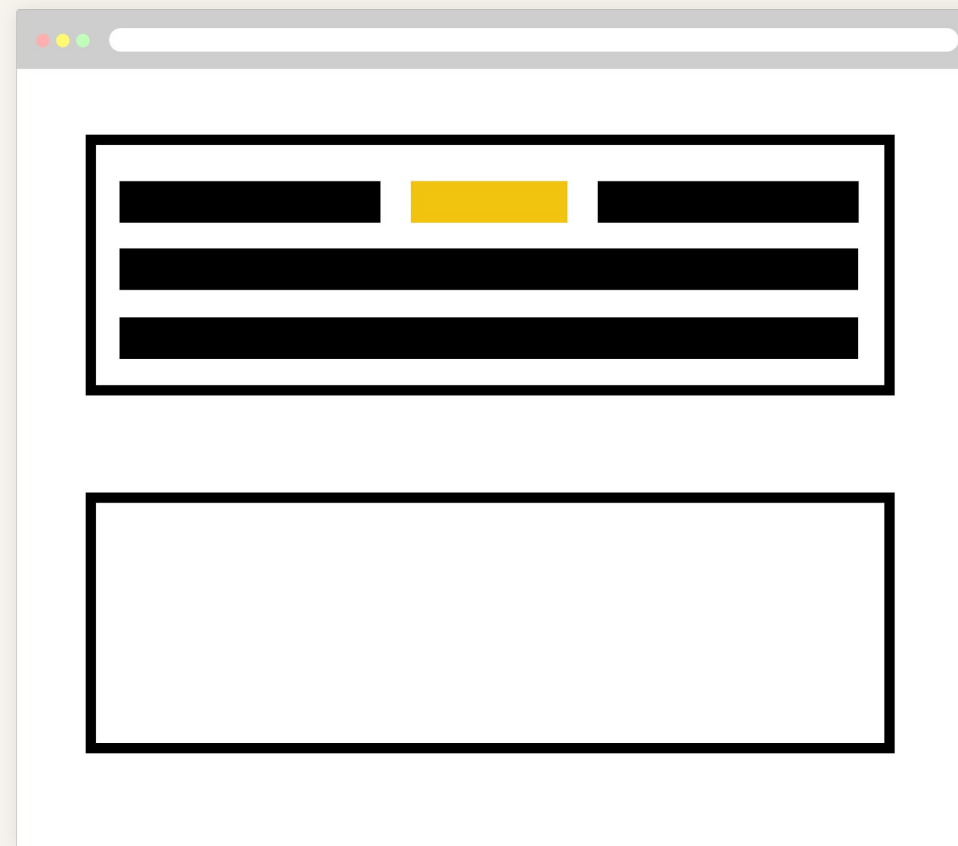


display: inline

This element takes up only as much width as it needs. Padding and margins only work for **left** and **right**, not top and bottom.

top and **bottom** spacing is controlled by the **line-height** property because the content is inline.

```
div {  
    display: inline;  
}
```

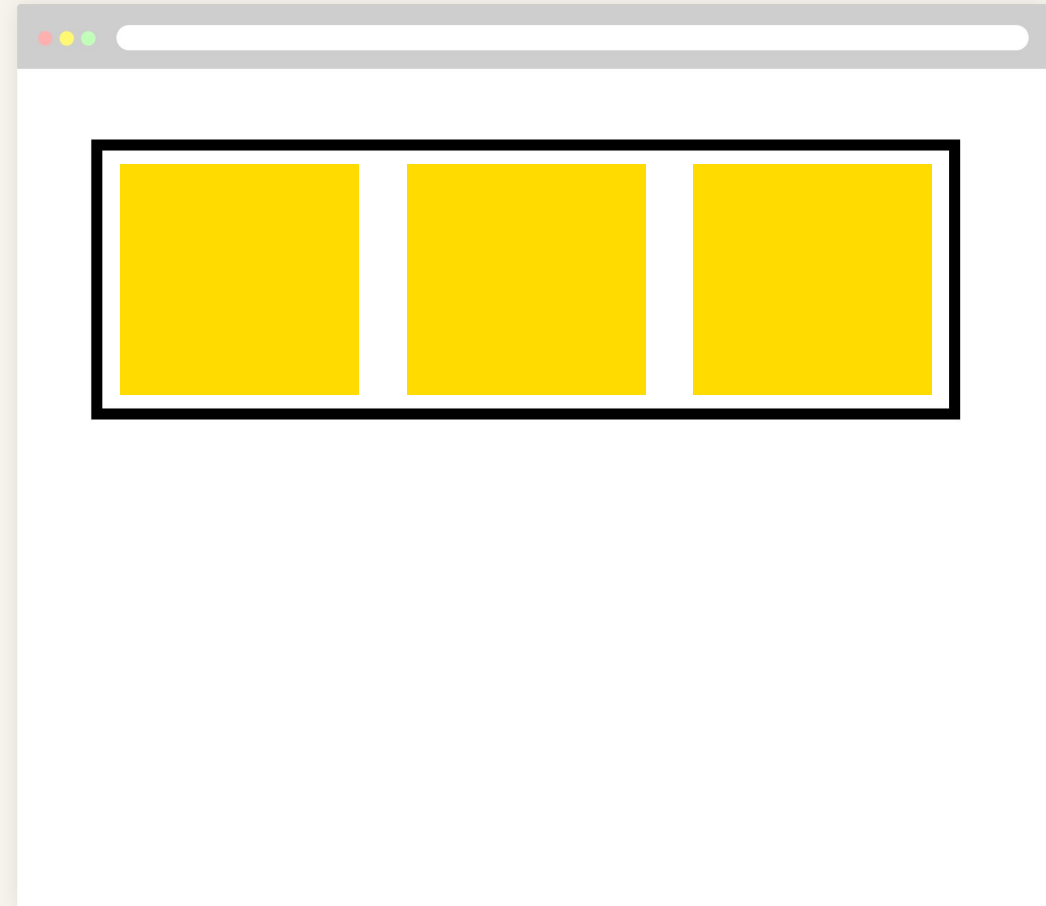


display: inline-block

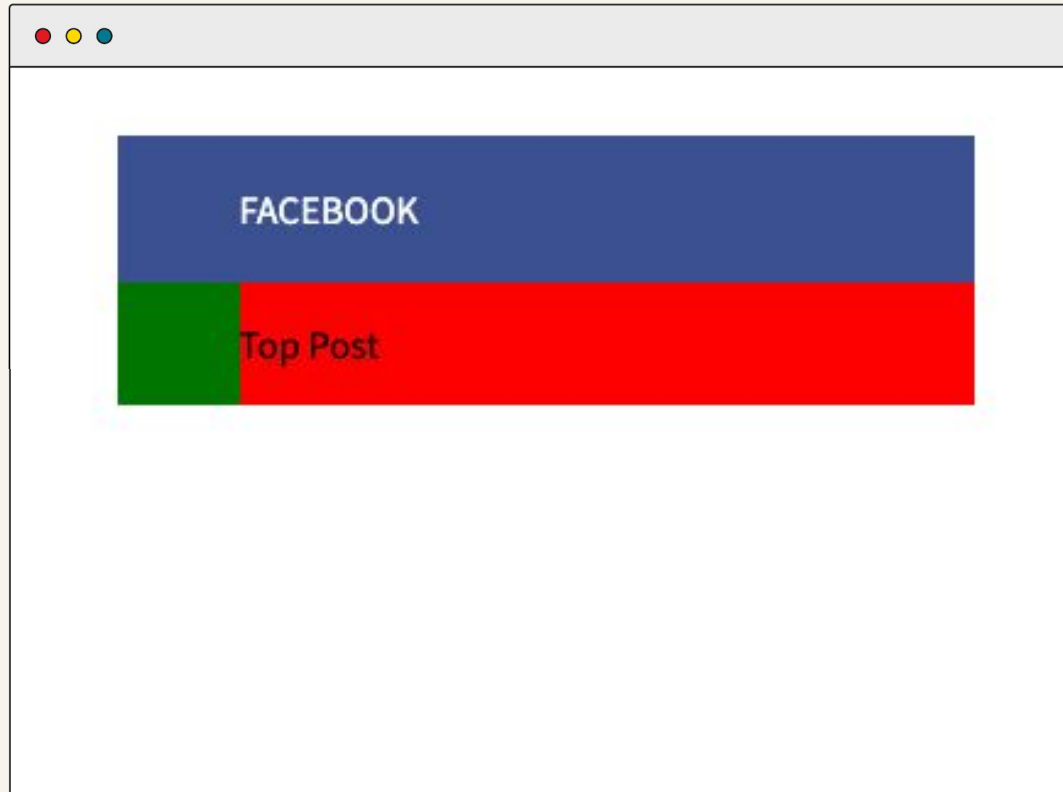
This combines the two concepts.

Inline blocks display inline with other items but allow you to use all **margin**, **padding**, **height**, and **width** properties.

```
div {  
    display: inline-block;  
}
```



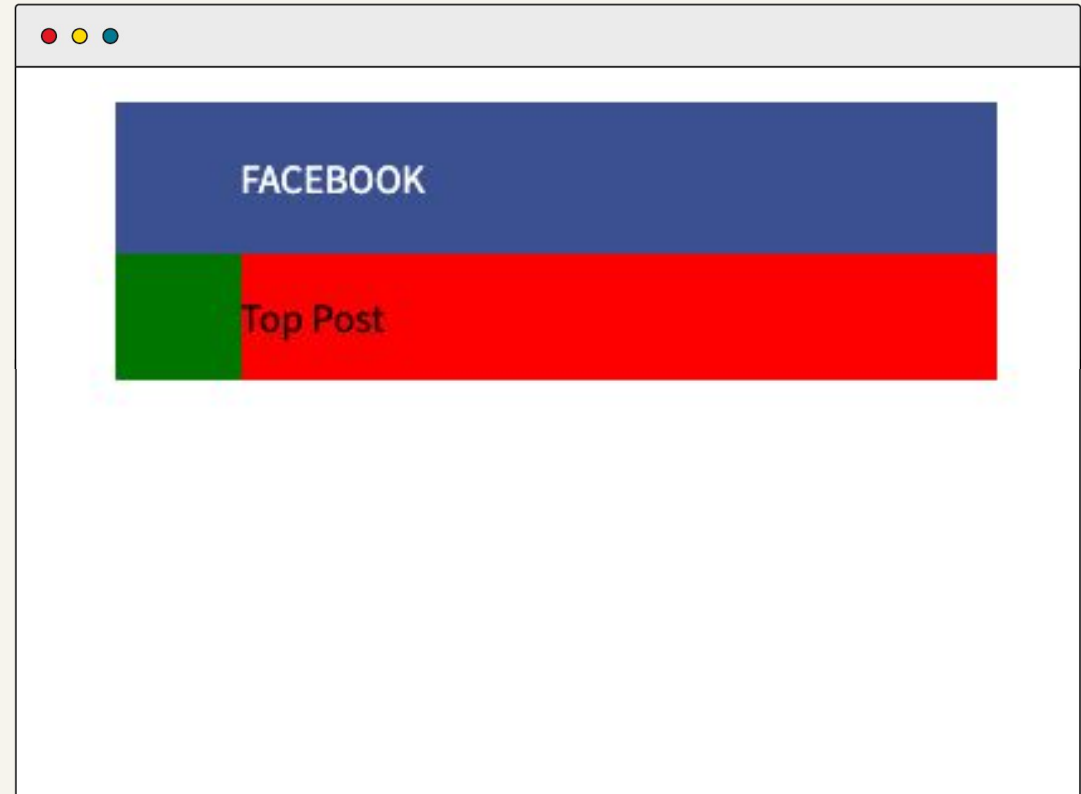
What next?



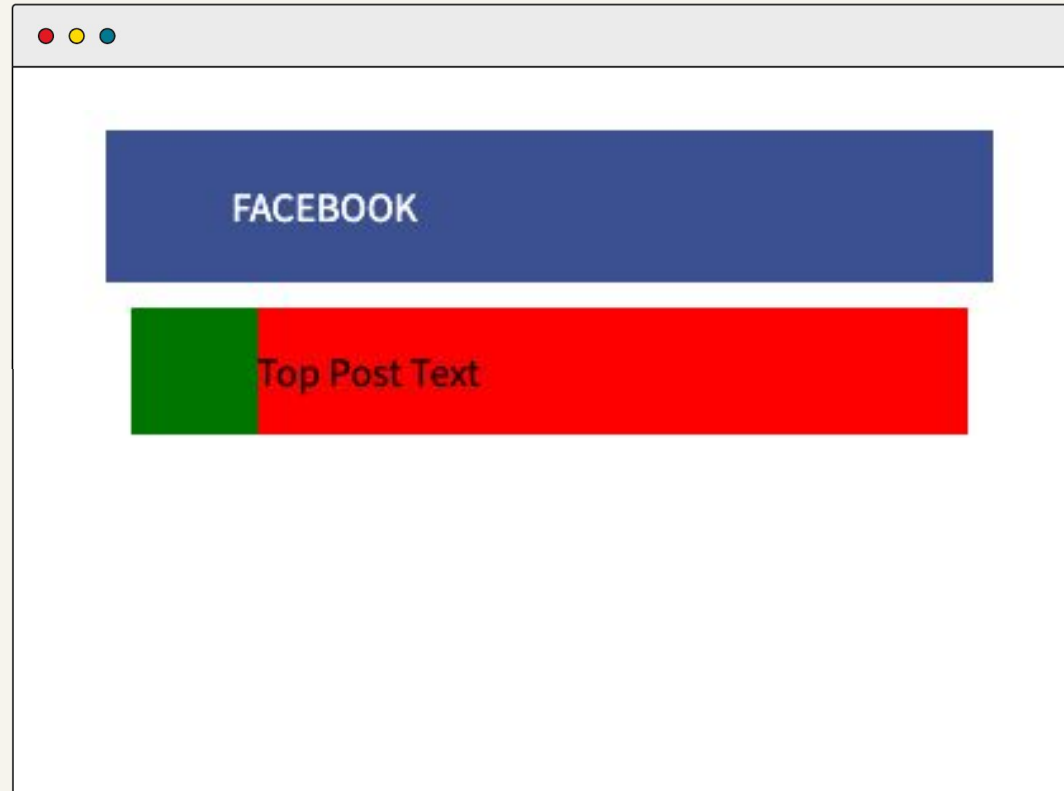
Add padding to the #top_post

```
<div id="top_post">  
  <div id="top_post_image"></div>  
  <div id="top_post_text">  
    Top Post Text  
  </div>  
</div>
```

```
#top_post {  
  padding: 10px;  
}
```

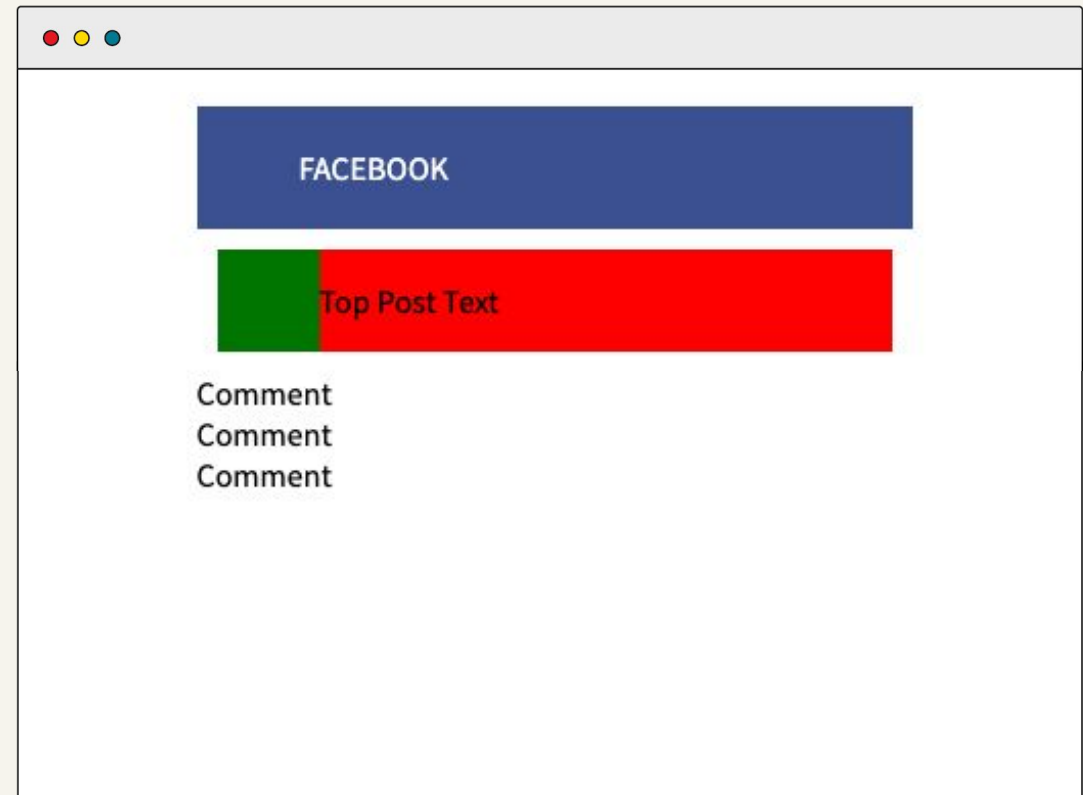


What next?

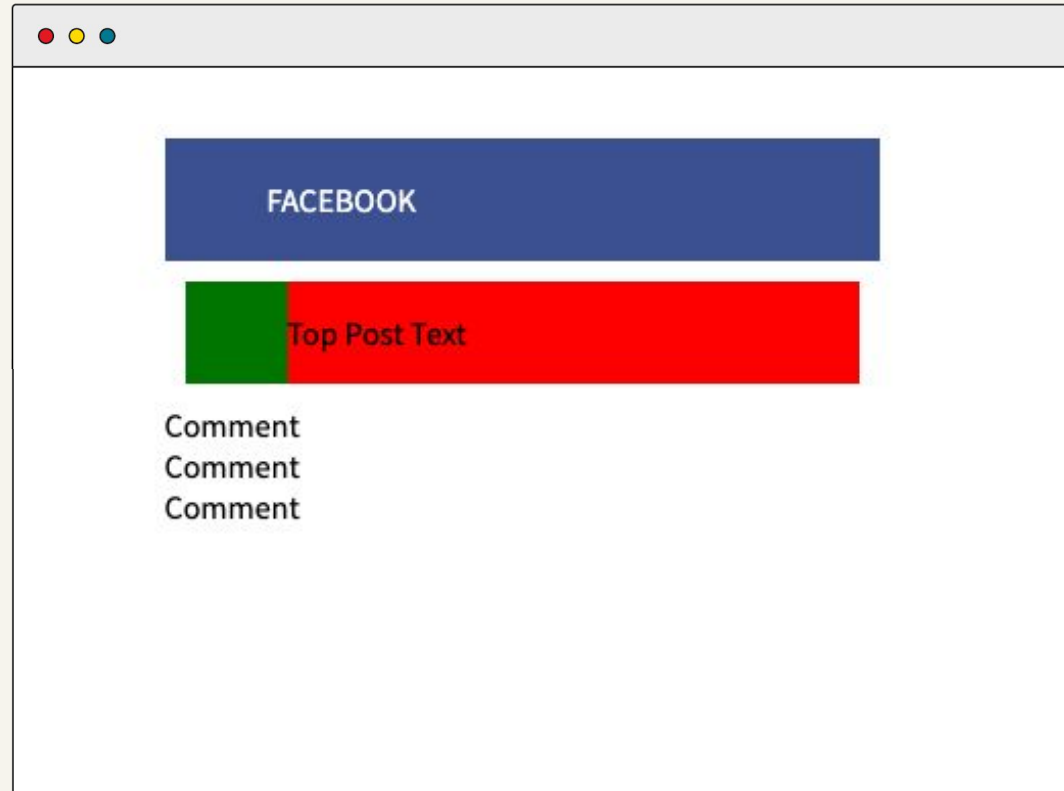


Add comments

```
<div id="comment_container">  
  <div class="comment">Comment</div>  
  <div class="comment">Comment</div>  
  <div class="comment">Comment</div>  
</div>
```



What next?

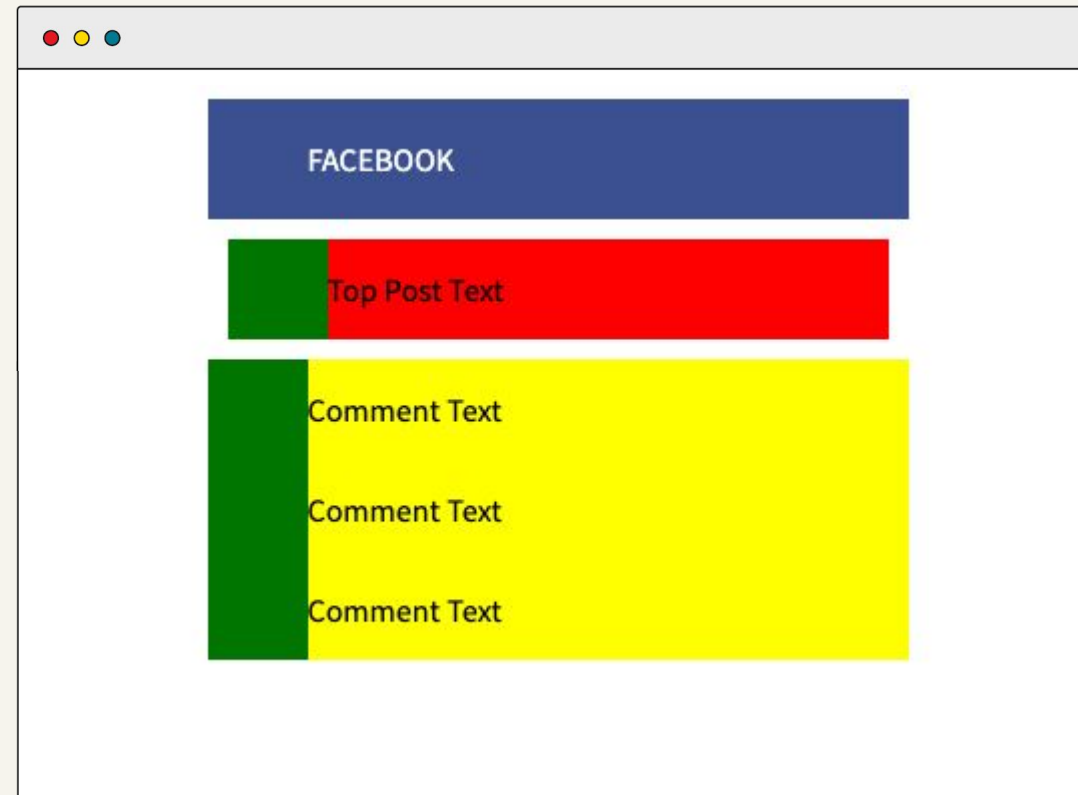


Comment substructure: image and text

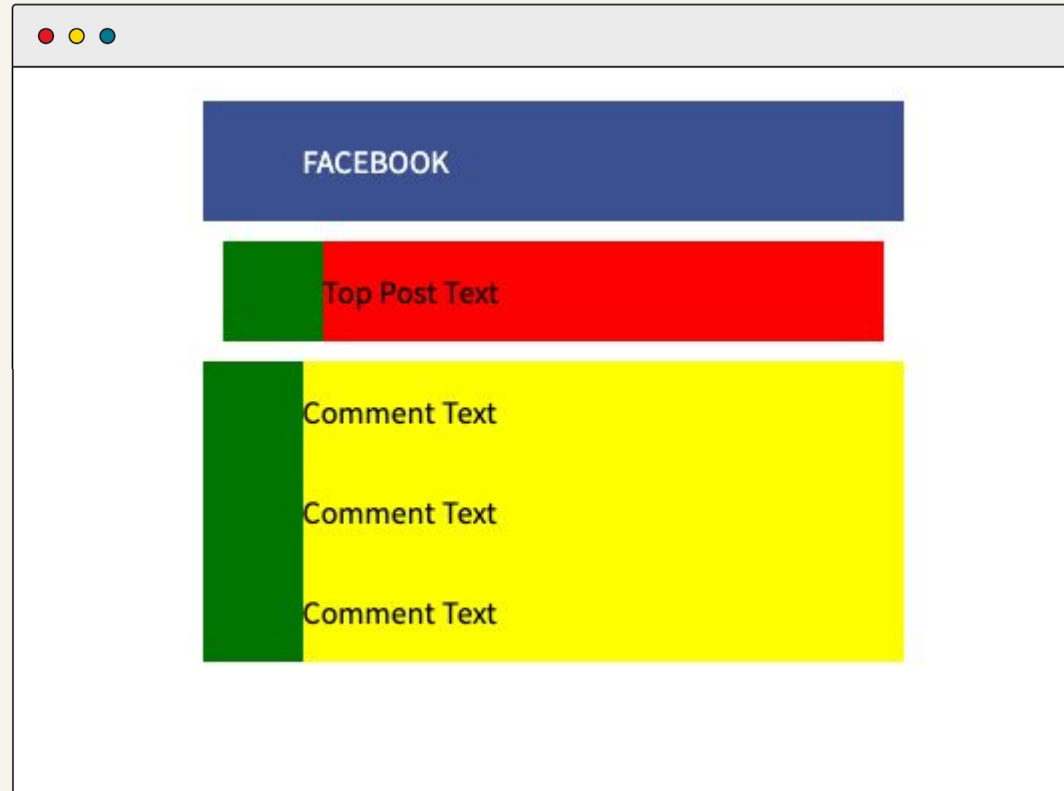
```
<div class="comment">
  <div class="comment_image"></div>
  <div class="comment_text">Comment Text</div>
</div>
```

```
.comment_image {
  height: 50px;
  width: 50px;
  background-color: green;
  display: inline-block;
}

.comment_text {
  height: 50px;
  width: calc(100% - 50px);
  display: inline-block;
  background-color: yellow;
  vertical-align: top;
  line-height: 50px;
}
```



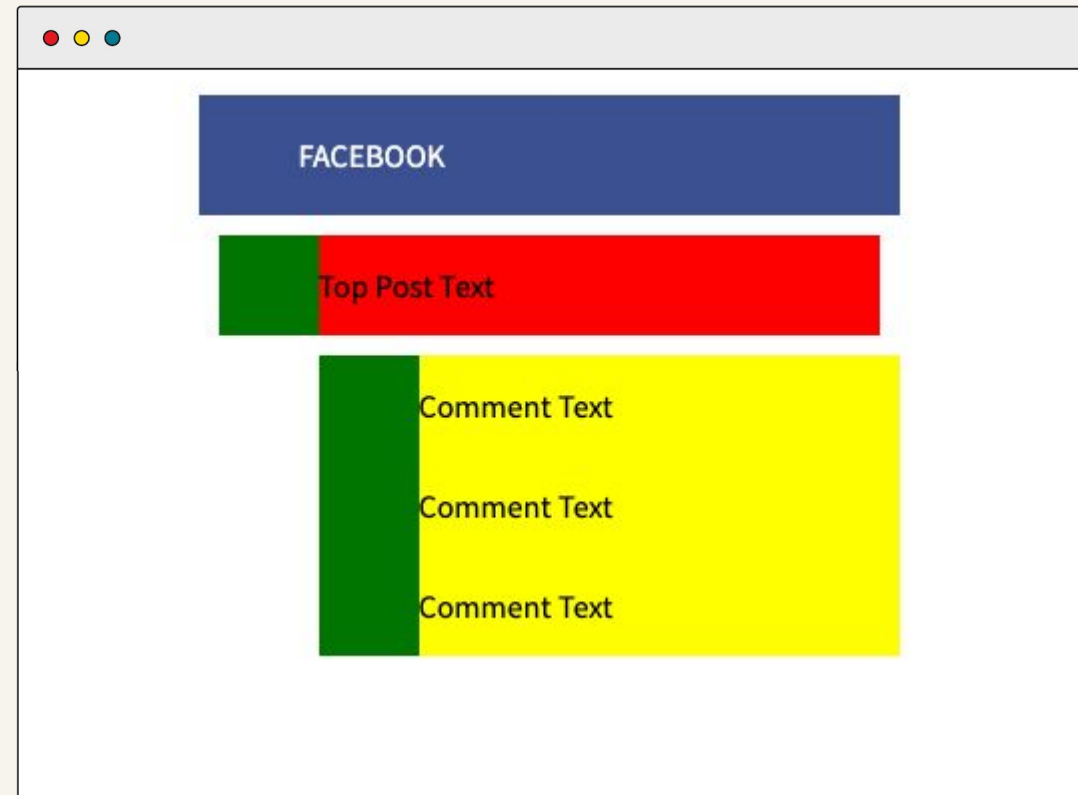
What next?



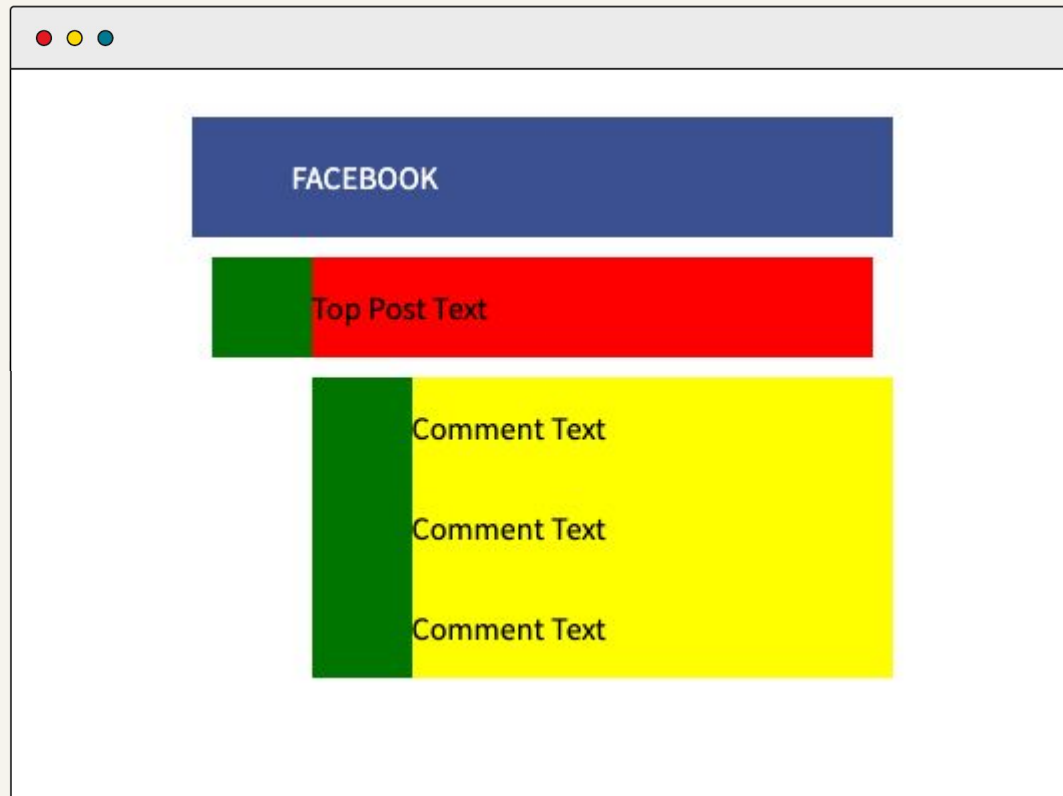
#comment_container positioning

```
<div id="comment_container">
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
</div>
```

```
#comment_container {
  margin-left: 60px;
}
```



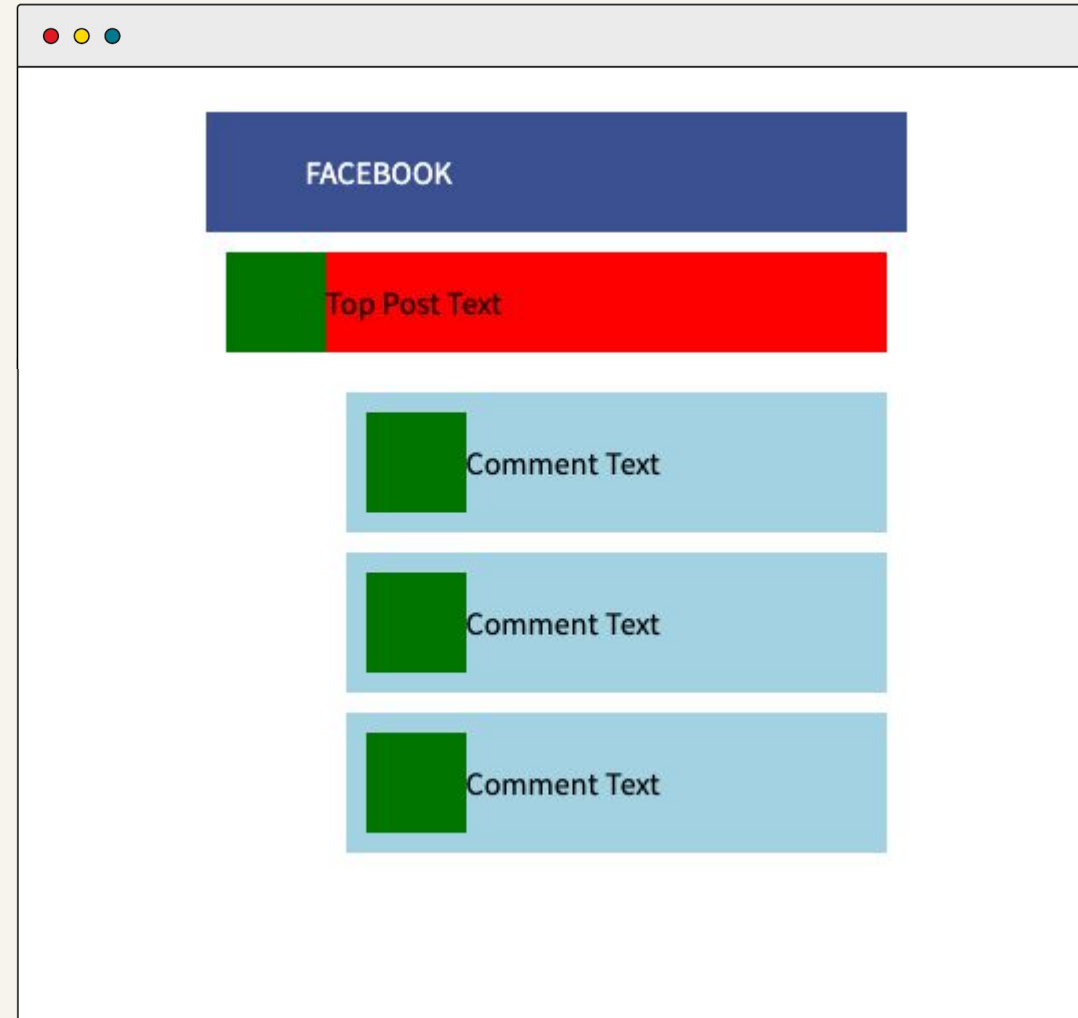
What next?



.comment class background-color

```
<div id="comment_container">
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
</div>
```

```
.comment {
  height: 50px;
  background-color: lightblue;
  padding: 10px;
  margin: 10px;
}
```

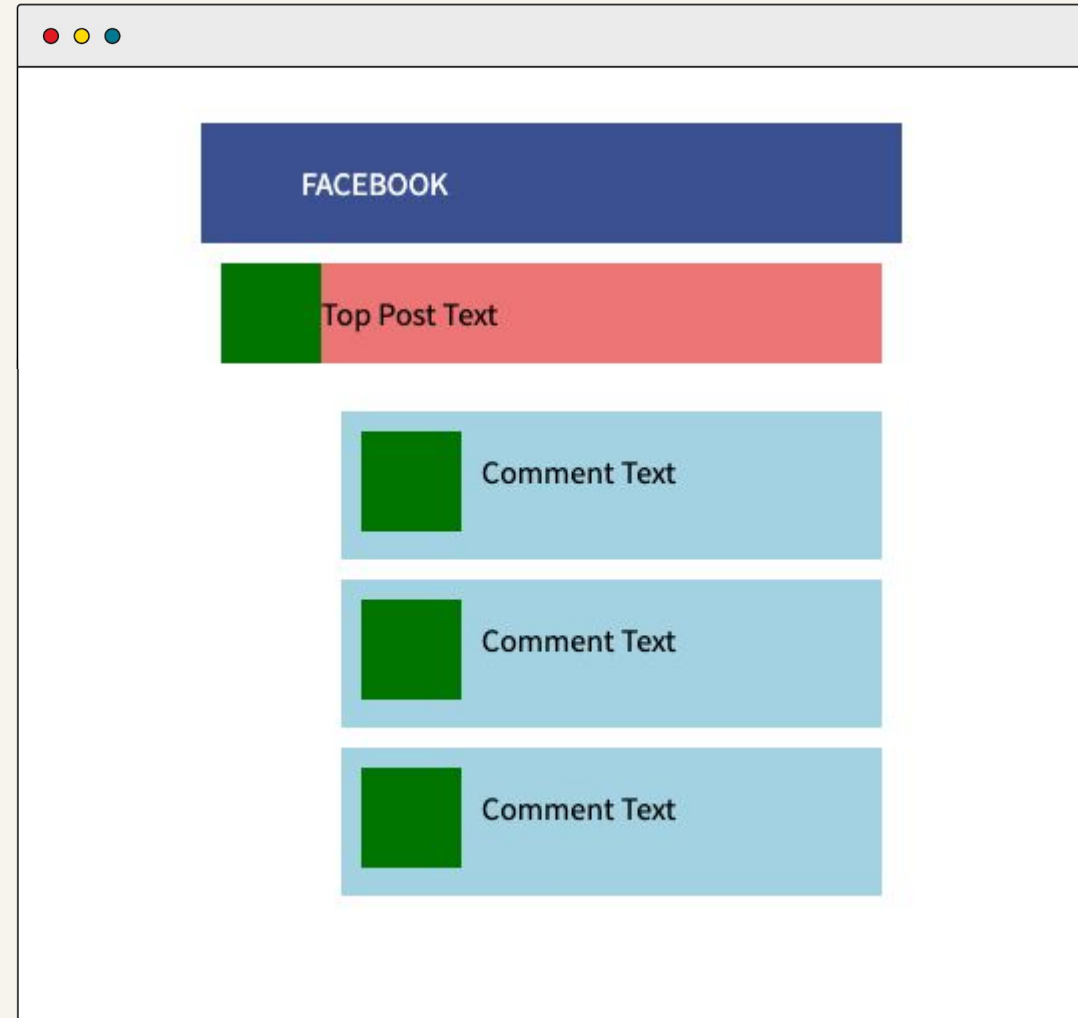


Reduce .comment_text width

```
<div id="comment_container">
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
  <div class="comment">
    <div class="comment_image"></div>
    <div class="comment_text">Comment Text</div>
  </div>
</div>
```

```
* {
  box-sizing: border-box;
}

.comment_text {
  height: 50px;
  width: calc(100% - 50px);
  display: inline-block;
  padding: 10px;
  vertical-align: top;
}
```



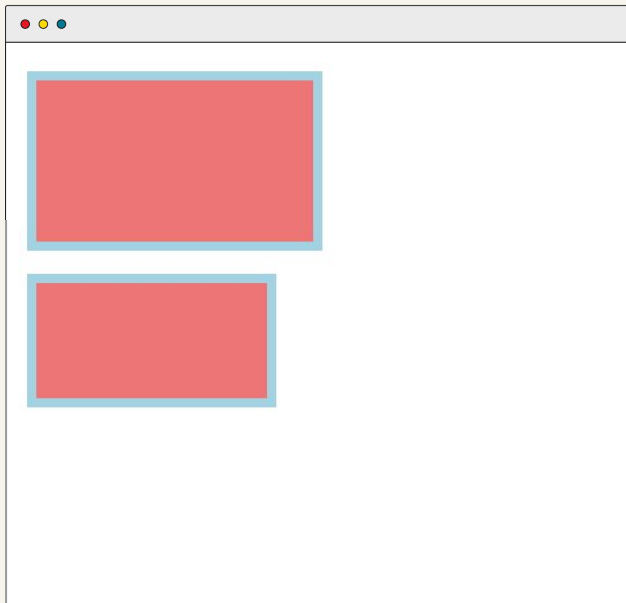
box-sizing

The CSS **box-sizing** property is used to adjust or control the size of any element that accepts a **width** or **height**. It specifies how to calculate the total width and height of that element.



without box-sizing

The CSS **box-sizing** property is used to adjust or control the size of any element that accepts a **width** or **height**. It specifies how to calculate the total width and height of that element.



```
<div class="box one"></div>
<div class="box two"></div>
```

.box .one

Total width: $200\text{px} + (2 * 20\text{px}) + (2 * 8\text{px}) = 256\text{px}$
Total height: $100\text{px} + (2 * 20\text{px}) + (2 * 8\text{px}) = 156\text{px}$

```
.box {
  width: 200px;
  height: 100px;
  border: 8px solid lightblue;
  background: lightcoral;
  margin-bottom: 20px;
}

.one {
  padding: 20px;
}
```



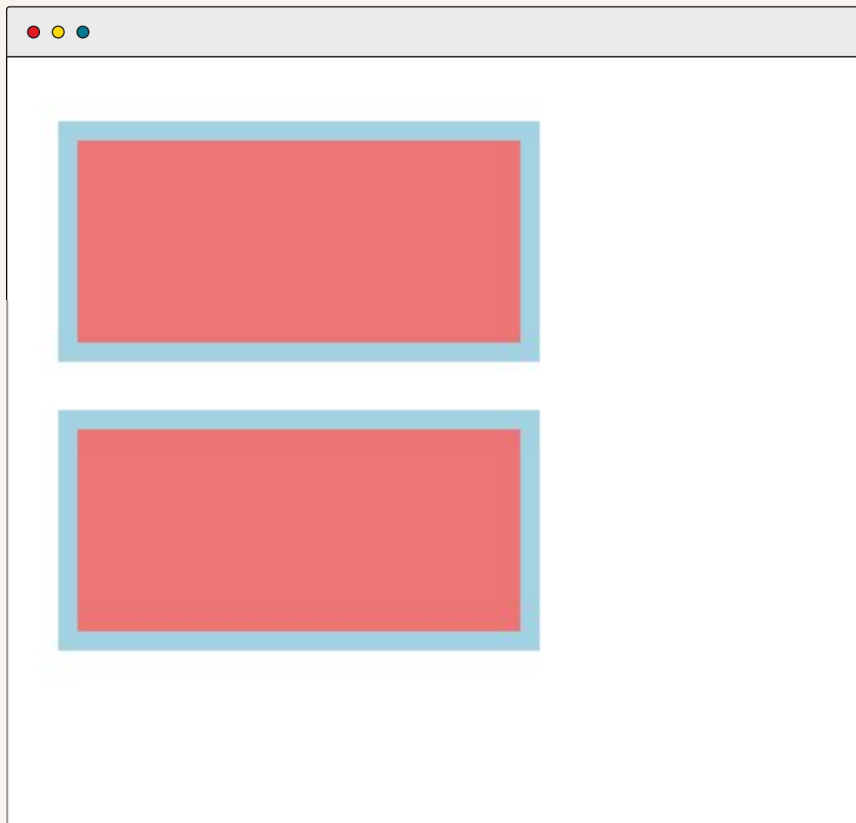
box-sizing: border-box

When you set the **box-sizing** property to **border-box**, it tells the browser to account for any **border** and **padding** assigned to the element's **width** and **height**.

That means, if you set an element's width to **200 pixels**, that 200 pixels will include any border or padding you add, and the content box will shrink to absorb that extra width.



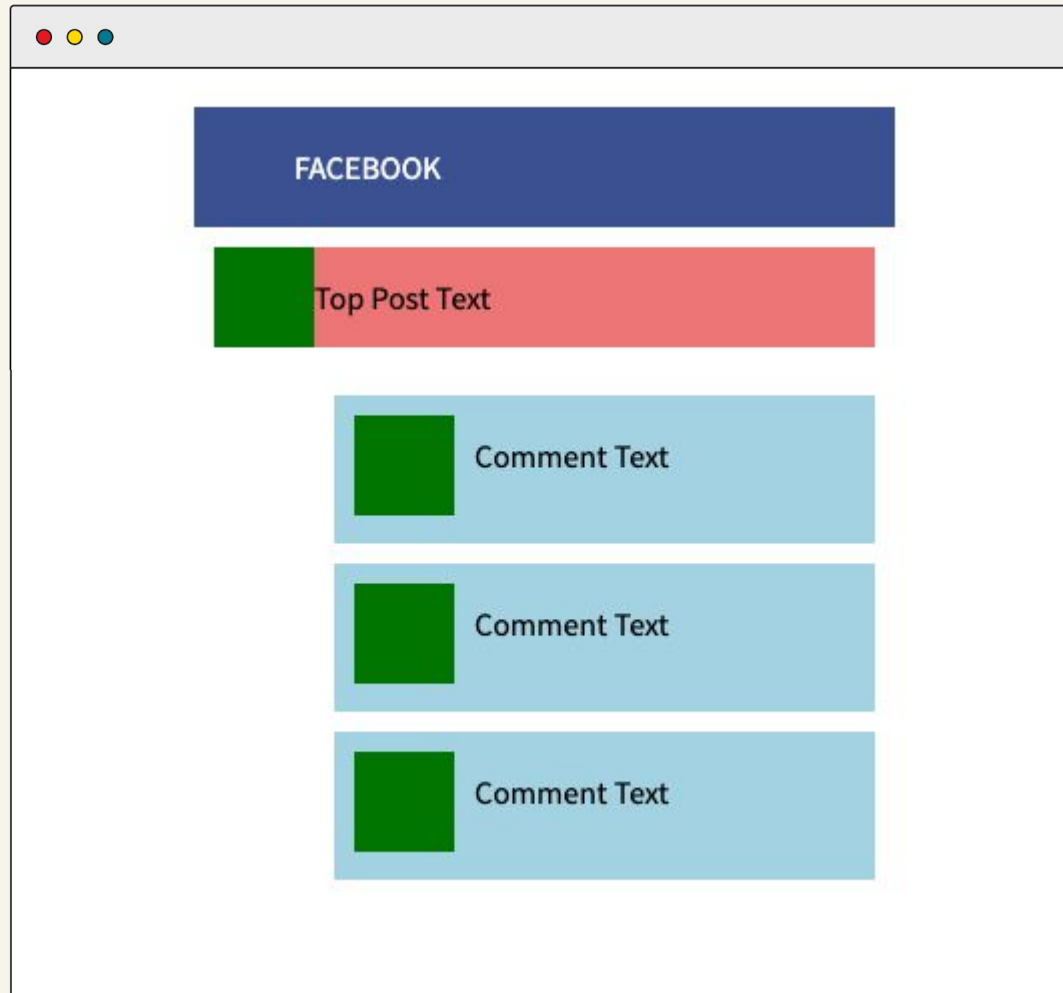
box-sizing: border-box



```
.box {  
  width: 200px;  
  height: 100px;  
  border: 8px solid lightblue;  
  background: lightcoral;  
  margin-bottom: 20px;  
}  
  
.one {  
  padding: 20px;  
  box-sizing: border-box;  
}  
  
.two {  
  box-sizing: border-box;  
}
```



What next? Am I done?



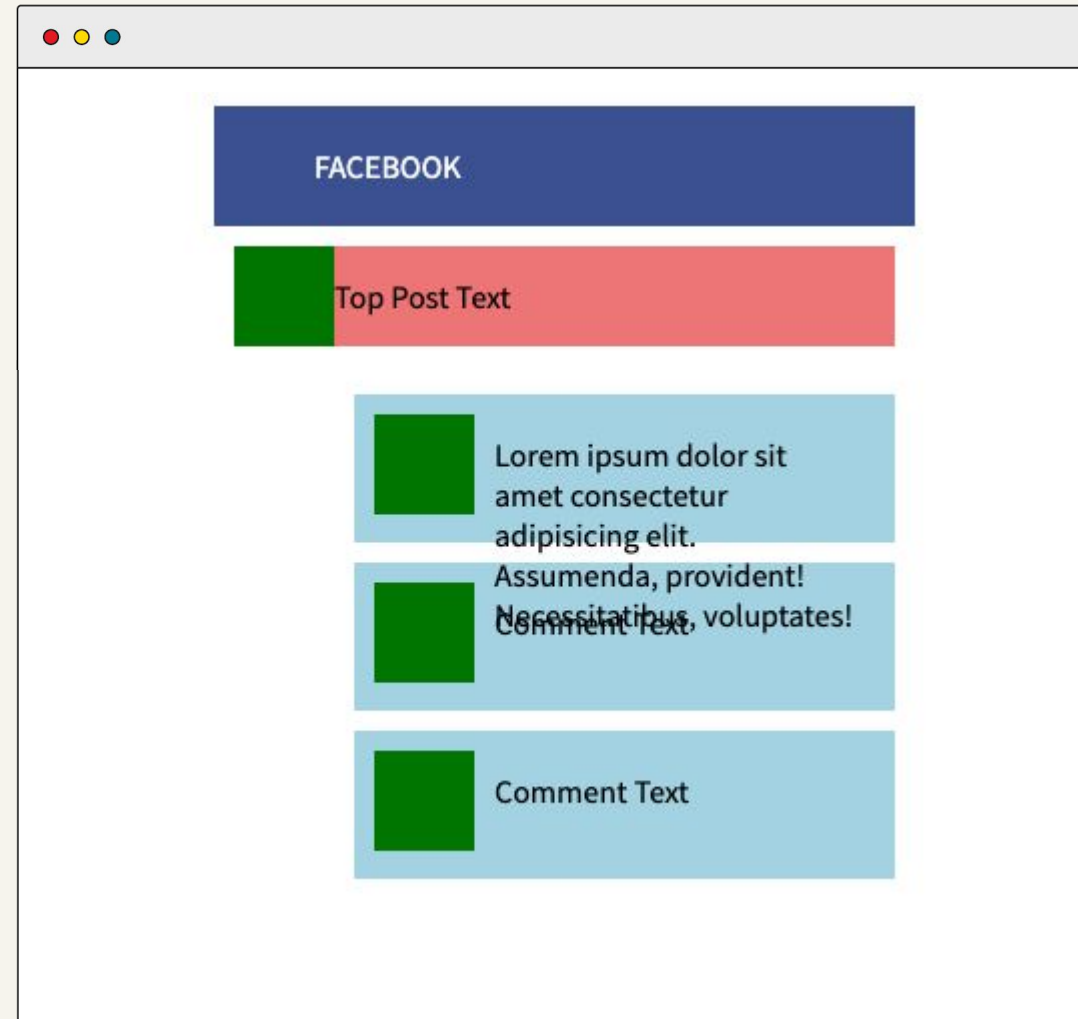
.comment_text is beyond boundaries of div



```
<div class="comment">
  <div class="comment_image"></div><!--
  --><div class="comment_text">Lorem ipsum dolor sit
  amet consectetur adipisicing elit. Assumenda,
  provident! Necessitatibus, voluptates!</div>
</div>
```



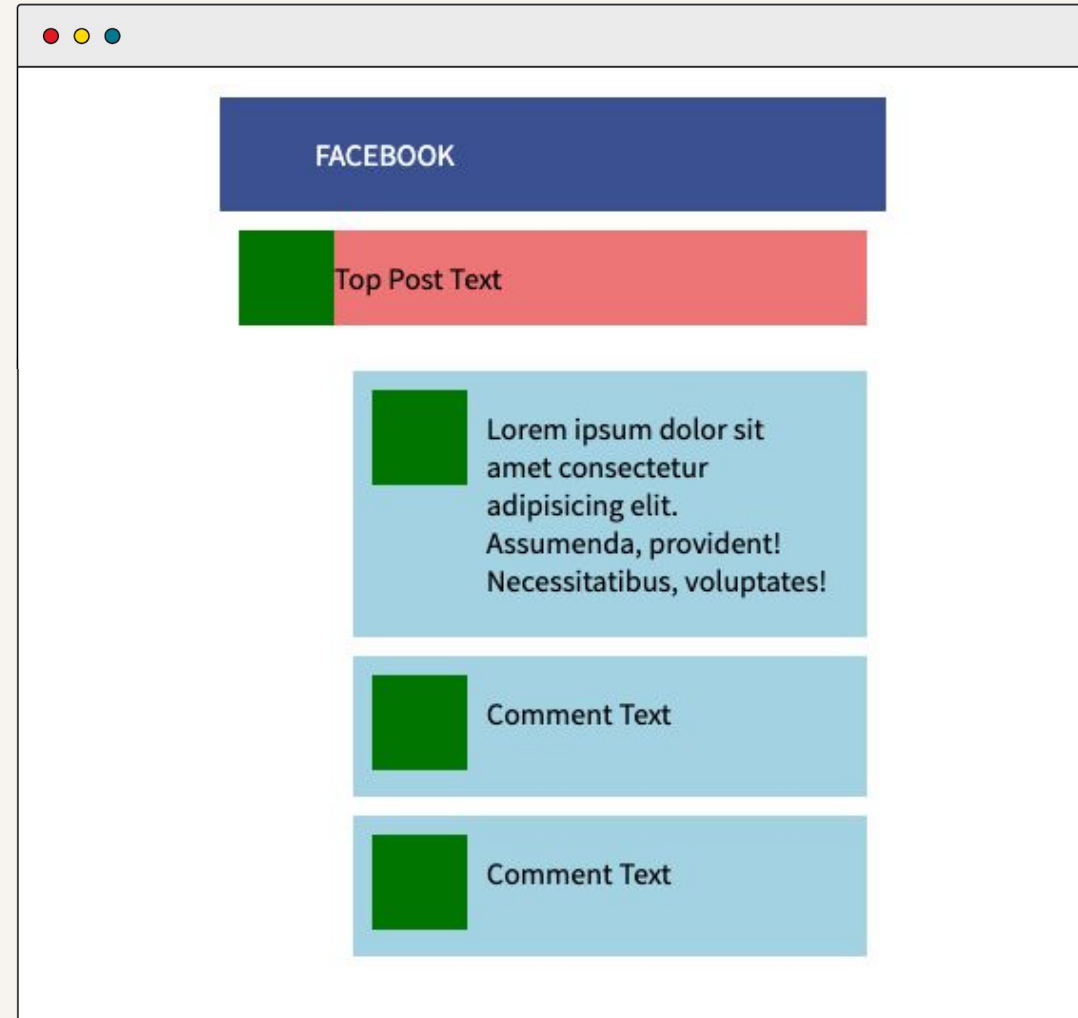
```
.comment_text {
  height: 50px;
  width: calc(100% - 50px);
  display: inline-block;
  padding: 10px;
  vertical-align: top;
}
```

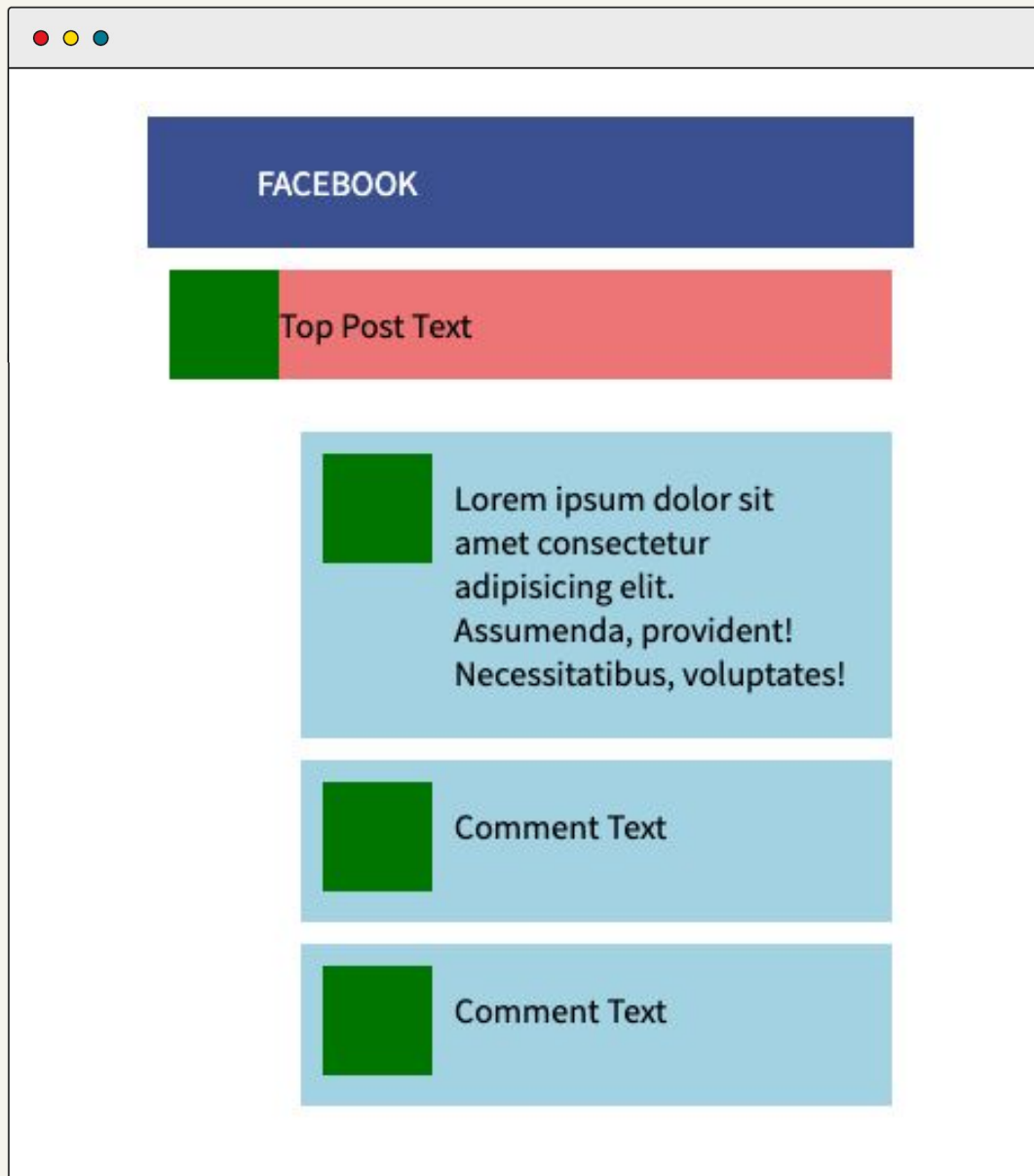


.comment_text height fixed

```
<div class="comment">
  <div class="comment_image"></div><!--
  --><div class="comment_text">Lorem ipsum dolor sit
  amet consectetur adipisicing elit. Assumenda,
  provident! Necessitatibus, voluptates!</div>
</div>
```

```
.comment_text {
  /* height: 50px; */
  width: calc(100% - 50px);
  display: inline-block;
  padding: 10px;
  vertical-align: top;
}
```



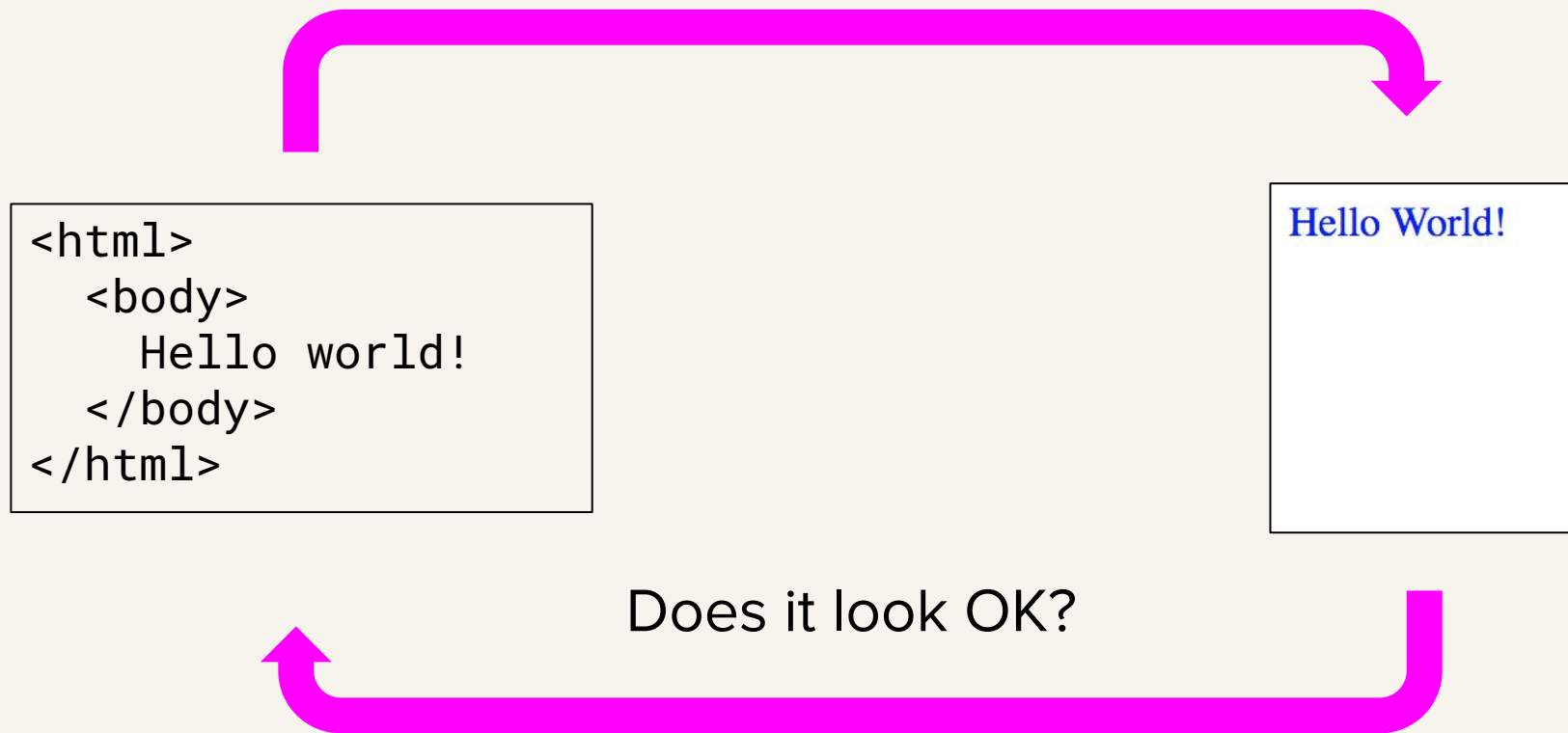


After layout...what next?

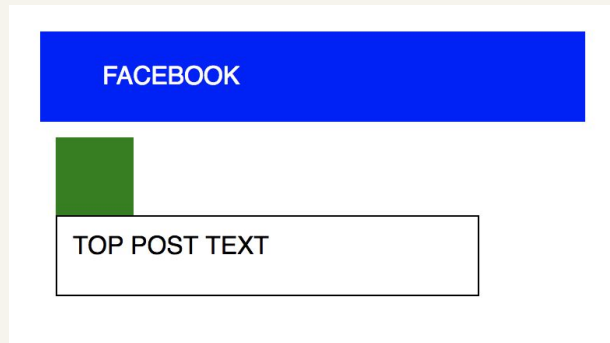


Iterative Style of Programming helps you build a mental model of your code.

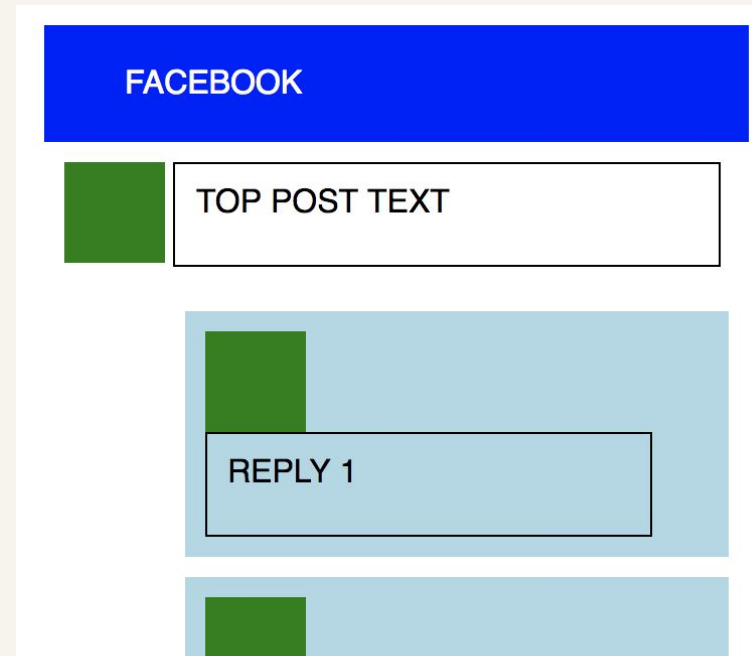
What is the **smallest** unit of progress I can make?



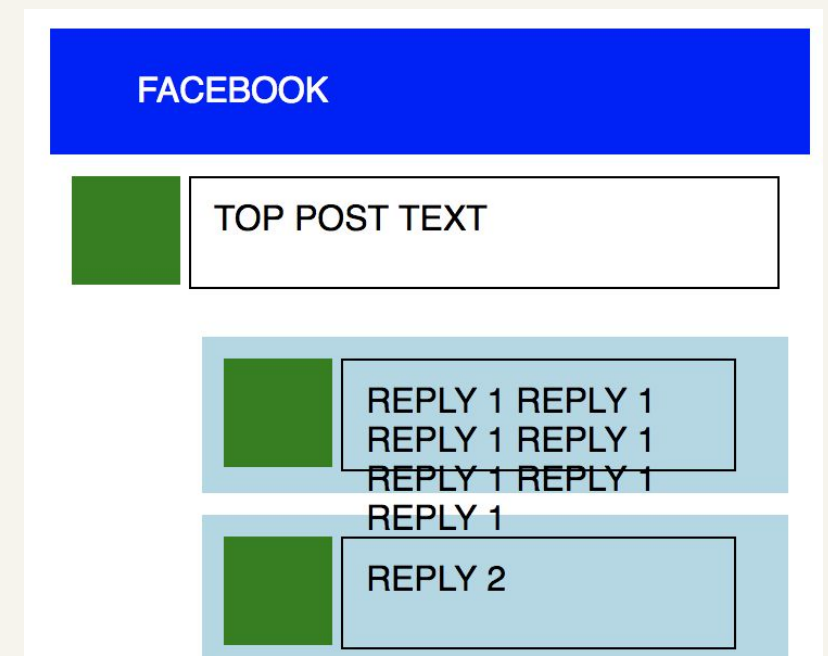
Iterative Style of Programming helped me recover from errors one at a time (as I created them)



Error 1
display: inline-block



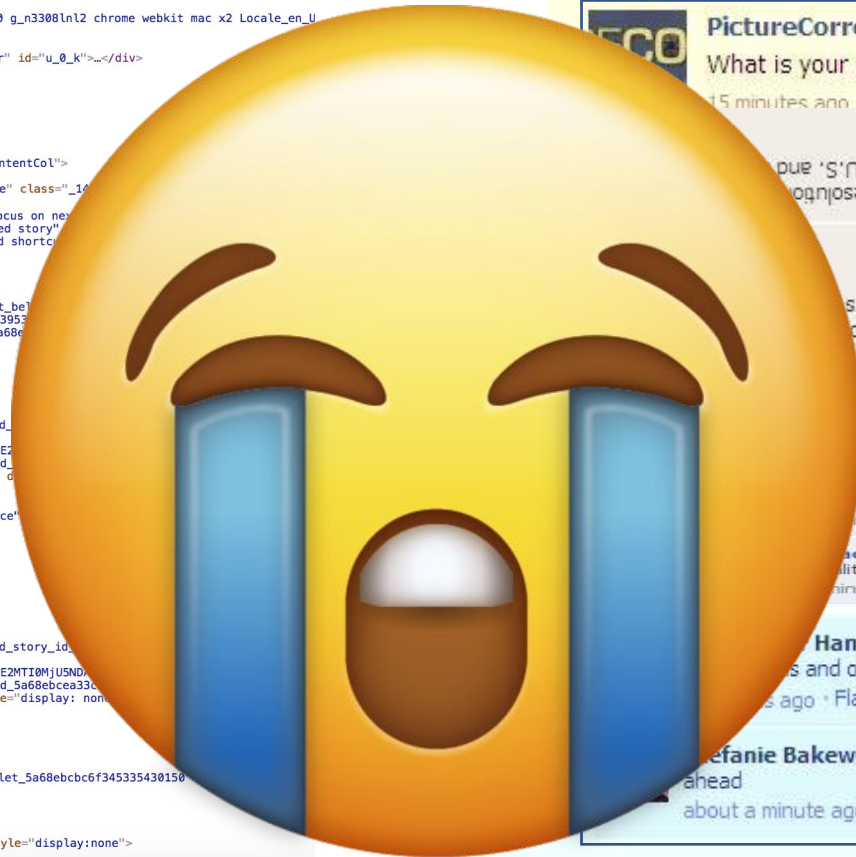
Error 2
.comment <div> padding



Error 3
.comment <div> height



**Once you have too many bugs,
you can't fix it – it's best to just start over**



Lecture 03: Grids and Layout

Grids

Currently, the dominant way to layout information



Our previous solution to the problem worked; but it would be much easier if we used a grid.



Mail ▾



More ▾

1-14 of 14



COMPOSE

Inbox (2)

Sent Mail

Drafts (54)

Need to Read

Technology Coach

More ▾



Primary



Social



Promotions



Updates



☐ ☆ > Baldwin, BPS (2)

Bloglovin' - Not that I know of. Sometimes blocked sites come in on a hug

11:08 am

☐ ☆ > Feldmann, Ann

Fwd: 21st Century Classroom Visit? - Ann Feldmann District Technolog

10:18 am

☐ ☆ > me, Lambert (2)

8th Period - Hey, Jeff. I am planning on doing the Mystery Hangout lesson

8:05 am

☐ ★ > me

(no subject) - -- Jeffrey Bernadt District Technology Specialist Bellevue Pu

Jan 12

☐ ★ > me, Braasch, Klammer (3)

Visitation & Observation Request - Good morning! The second semester is

Jan 12

☐ ★ > me, Cox (2)

Bri Cox - Coaching Doc - Invitation to edit - Jeff, I finally was able to read a

Jan 12

☐ ★ > Tripple, Todd

3D Printer - Hello Amy, I received some information from a 3D printing exp

Jan 11

☐ ★ > donotreply

Education Preview Build Notice - AppleSeed Build Notice Education Pre

Jan 11

☐ ★ > Ann Feldmann

Fwd: [New post] What "Innovative Leadership" Looks Like - A good read! :

Jan 11

☐ ★ > me, Campbell (7)

Khan Academy Issues - It kept repeating that "changed password 7 month

Jan 8

☐ ☆ > Toelle, me (3)

6 Word Stories - Hey Jeff! We just returned from lunch, but still have quite

Jan 8

☐ ★ > me, Evon (4)

Request to Visit on Thursday AM - Hi Jeff, Happy New Year! We would lov

Jan 6

☐ ☆ > Jeffrey Bernadt

Shared from Twitter: What teachers need to know about multicultural educ

12/23/15



Search people...

● Jed Givens

● Ann Feldmann

Ann Feldmann

Cadwallader, Julie

Christopher Kessell

Courtney Saffold

Jeanette Carlson

Johnson, Stephanie

Kelly Bernadt

Loomis, Phillip

Robert Mason

Shannon Gier

Todd Tripple

Toledo, Julian (Delor...





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216,782 views



2K



48



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Would I Lie to You? - S11 E08 (NEW WILTY 19th January)

David Mitchell

Recommended for you

New



Actor Replacements That Totally Ruined The Show

Looper

3M views



Inside Matthew Perry's Tragic Real Life Story

Nicki Swift

952K views



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Favorite Videos

1M views



REACT S7 • E19

TEENS REACT TO WINDOWS 95

FBE

12M views



Friends - Ross and Monica's Cousin

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Lecture 03: Grids and Layout

Twitter Bootstrap

An implementation of grid layout





Build fast, responsive sites with Bootstrap

Quickly design and customize responsive mobile-first sites with Bootstrap, the world's most popular front-end open source toolkit, featuring Sass variables and mixins, responsive grid system, extensive prebuilt components, and powerful JavaScript plugins.

[Get started](#)[Download](#)

Currently **v5.0.0-beta1** · [Previous releases](#)



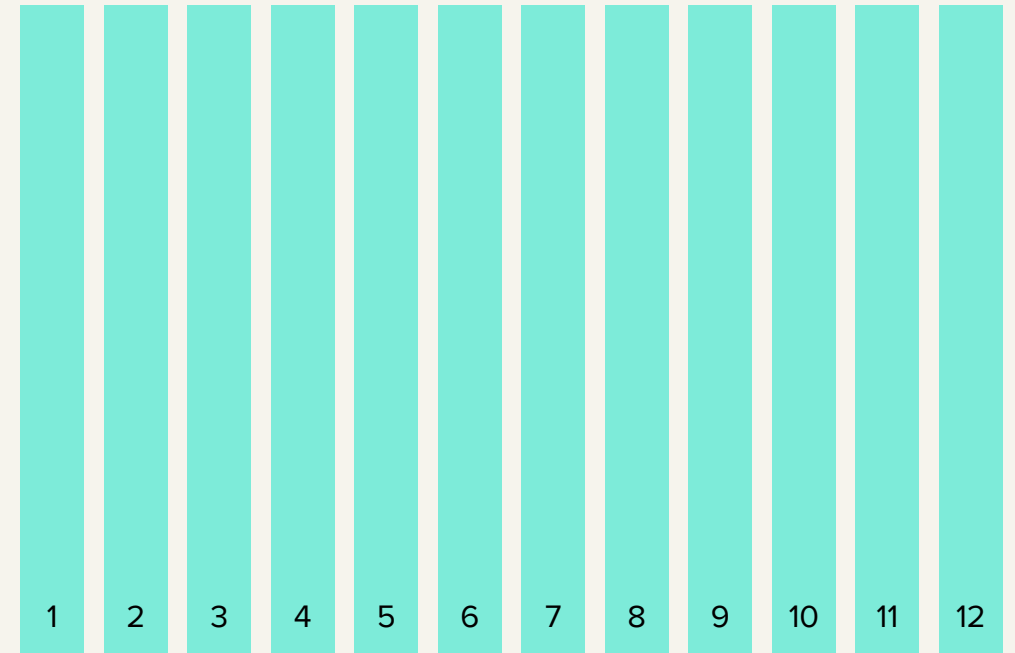
Easily add AI to your apps with an Azure free account.

adrian carlson

73 • [Spring 2026](#) • [User Interface Design](#)

Grid-Based Design

- Bootstrap sets up a series of 12 columns that automatically respond to screen size.



How many columns do you see?

Columbia University

@Columbia

"The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President

New York, New York

columbia.edu

Joined February 2011

Tweet to Columbia University

4 Followers you know



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Tweets Tweets & replies Media



Columbia University @Columbia · 39m
Groundbreaking. Beautiful. Monumental. Posing Modernity: The Black Model from Manet and Matisse to Today is closing soon. Come now, it's free!
[@ColumbiaSOA](#) #arts #Harlem [ow.ly/Q8rM30nssdJ](#)



Who to follow · Refresh · View all



Princeton University ...

Follow



Yale University @Yale

Follow



Harvard University @H...

Follow



Find people you know
Import your contacts from Gmail

Connect other address books

Trends for you · Change

#SDLive
37.9K Tweets

#Sabres
2,607 Tweets

James Ingram
R&B legend James Ingram has died at age 66

Rusev
3,867 Tweets

#SBGC

How wide is the middle? (compared to the sides)

Columbia University

@Columbia

"The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President

New York, New York

columbia.edu

Joined February 2011

Tweet to Columbia University

4 Followers you know



2,754 Photos and videos



Tweets

Tweets & replies

Media



Columbia University @Columbia · 39m

Groundbreaking. Beautiful. Monumental. Posing Modernity: The Black Model from Manet and Matisse to Today is closing soon. Come now, it's free!

@ColumbiaSOA #arts #Harlem [ow.1/Q8rM30nssdJ](#)



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Yale University @Yale

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Harvard University @H...

Follow



Find people you know

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[Connect other address books](#)

Trends for you · [Change](#)

#SDLive

37.9K Tweets

#Sabres

2,607 Tweets

James Ingram

R&B legend James Ingram has died at age 66

Rusev

3,867 Tweets

#SBGC

All content goes in the main container



The container is divided into 12 columns



You get to apportion those 12 columns

Columbia University

@Columbia

"The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President

New York, New York

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Tweets

Tweets & replies

Media



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#SDLive

37.9K Tweets

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James Ingram

R&B legend James Ingram has died at age 66

Rusev

3,867 Tweets

#SBGC

3,521 Tweets

Twelve columns split into 3-6-3

Columbia University

@Columbia

"The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President

New York, New York

columbia.edu

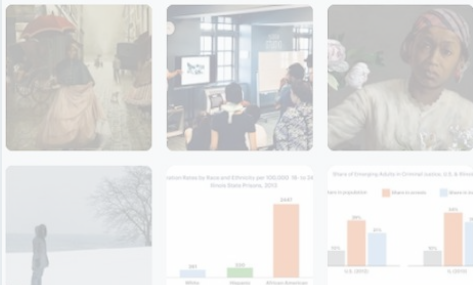
Joined February 2011

Tweet to Columbia University

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2,754 Photos and videos



Tweets

Tweets & replies

Media



Columbia University @Columbia · 39m

Groundbreaking. Beautiful. Monumental. Posing Modernity: The Black Model from Manet and Matisse to Today is closing soon. Come now, it's free!

@ColumbiaSOA #arts #Harlem ow.ly/Q8rM30nssdJ



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Yale University @Yale

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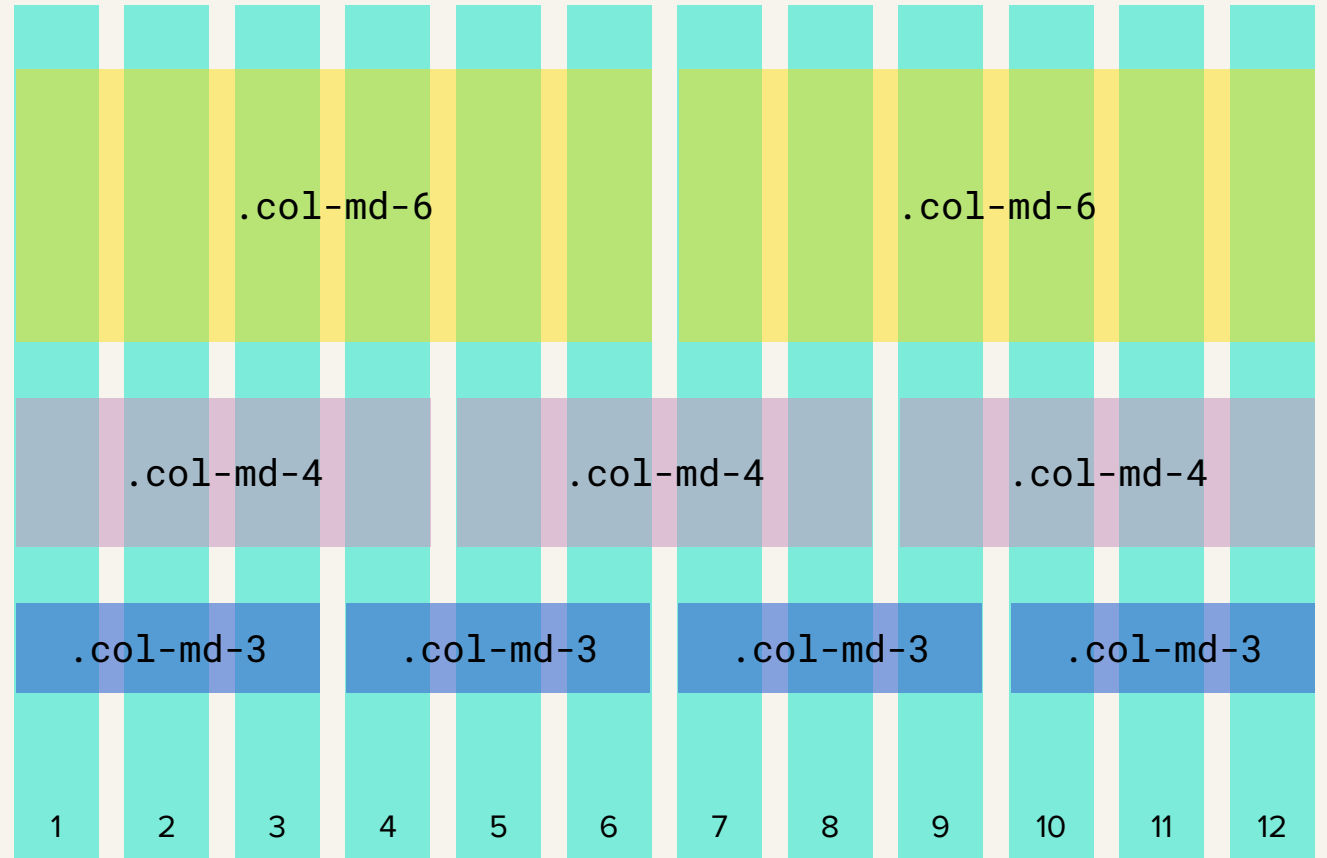
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Bootstrap Grid

- You can create boxes that clip into the grid and take up 1–12 columns.



Bootstrap Grid

All content should be placed within grid columns. Build column classes using the following formula:

col-{screen-size}-{width}

Examples of Bootstrap column classes:



Do the columns have the same rows?

@Columbia

📍 New York, New York

 columbia.edu

 Joined February 2011

Tweet to Columbia University

 4 Followers you know

 2,754 Photos and videos

Tweets & replies

Columbia University  @Columbia · 39m

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Is there alignment within the column?

Columbia University

@Columbia

"The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President

New York, New York

columbia.edu

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#SBGC

Can you use the original grid? ... no

A screenshot of a Twitter profile for Columbia University. The profile header shows the name "Columbia University" with a verified badge and the handle "@Columbia". The bio reads: "The best education is one that prepares you for your own venture into the unknown." - Lee C. Bollinger, Columbia University President. The location is "New York, New York" and the website is "columbia.edu". It was joined in February 2011. The tweet section shows a tweet from 39 minutes ago about a painting titled "Posing Modernity: The Black Model from Manet and Matisse to Today" which is closing soon. The tweet includes a large image of a painting depicting a woman in a blue coat and a long, patterned skirt holding a red umbrella on a cobblestone street. The right sidebar shows a "Who to follow" section with Princeton University, Yale University, and Harvard University. Below that is a "Trends for you" section with hashtags #SDLive, #Sabres, and mentions of James Ingram and Rusev. A red grid is overlaid on the entire image, with a vertical orange line on the left side.

Grids within grids

Tweets

Tweets & replies

Media

Columbia University @Columbia · 3m
Groundbreaking. Beautiful. Monumental. Posing Modernity: The Black Model from Manet and Matisse to Today is closing soon. Come now, it's free! @ColumbiaSOA #arts #Harlem [now.ly/Q8M30nssdJ](#)

ice Design



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Follow



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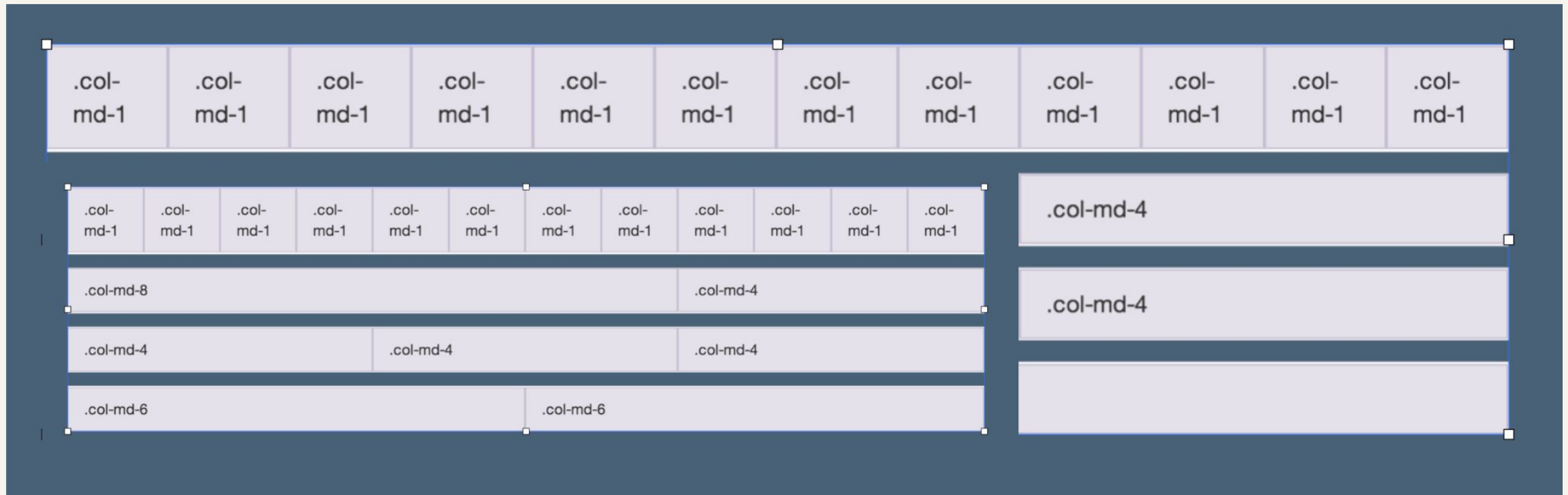
3,867 Tweets

#SBGC

3,521 Tweets

Bootstrap (nested) grid.

Apportion 12 columns within a column.





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Moments · View all

Isaiah Thomas becomes a meme...

NBA · 2 hours ago



This bot-generated Coachella...

Haha · 3 hours ago



People are going crazy over this...

[Business Insider](#) · 5 hours ago

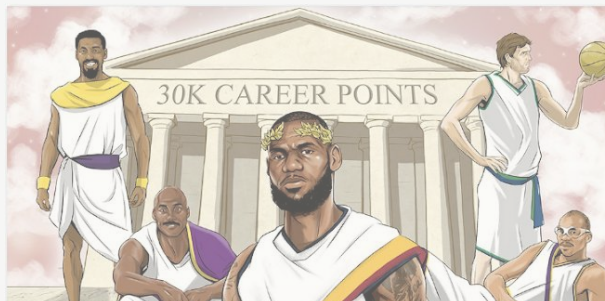


New to Twitter?

Sign up now to get your own personalized timeline!

[Sign up](#)

Featured Tweets

[General News ▸](#)[Sports commentary ▸](#)[World News ▸](#)

Add Bootstrap to your HTML

Place Bootstrap's stylesheet `<link>` in the `<head>` tag **before** your own custom stylesheets.

Many components require the use of JavaScript to function. Place the `<script>` near the end of your page, right before the closing `</body>` tag, to enable it.

```
<!doctype html>
<html lang="en">
  <head>
    <!-- Required meta tags -->
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">

    <!-- Bootstrap CSS -->
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta1/dist/css/bootstrap.min.

  <title>Hello, world!</title>
</head>
<body>
  <h1>Hello, world!</h1>

  <!-- Optional JavaScript; choose one of the two! -->

  <!-- Option 1: Bootstrap Bundle with Popper -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta1/dist/js/bootstrap.bund

  <!-- Option 2: Separate Popper and Bootstrap JS -->
  <!--
  <script src="https://cdn.jsdelivr.net/npm/@popperjs/core@2.5.4/dist/umd/popper.min.js"
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.0.0-beta1/dist/js/bootstrap.min.
  -->
</body>
</html>
```

Copy

Bootstrap requires a containing element to wrap site contents and house our grid system.

```
<div class="container">  
  ...  
</div>
```

Use `.container` for a responsive fixed width container.

```
<div class="container-fluid">  
  ...  
</div>
```

Use `.container-fluid` for a full width container, spanning the entire width of your viewport.



Add rows

```
<div class="container">  
  <div class="row">  
    ...  
  </div>  
</div>
```

Use rows to create horizontal groups of columns.

Only columns may be immediate children of rows.



Add columns

```
<div class="container">
  <div class="row">
    <div class="col-md-5">.col-md-5</div>
    <div class="col-md-7">.col-md-7</div>
  </div>
  <div class="row">
    <div class="col-md-4">.col-md-4</div>
    <div class="col-md-4">.col-md-4</div>
    <div class="col-md-4">.col-md-4</div>
  </div>
</div>
```



Breakpoints

Grid options

Bootstrap's grid system can adapt across all six default breakpoints, and any breakpoints you customize. The six default grid tiers are as follow:

- Extra small (xs)
- Small (sm)
- Medium (md)
- Large (lg)
- Extra large (xl)
- Extra extra large (xxl)

As noted above, each of these breakpoints have their own container, unique class prefix, and modifiers. Here's how the grid changes across these breakpoints:

	xs <576px	sm ≥576px	md ≥768px	lg ≥992px	xl ≥1200px	xxl ≥1400px
Container max-width	None (auto)	540px	720px	960px	1140px	1320px
Class prefix	.col-	.col-sm-	.col-md-	.col-lg-	.col-xl-	.col-xxl-
# of columns	12					
Gutter width	1.5rem (.75rem on left and right)					

All content should be placed within grid columns.

Build column classes using the following formula:

col-{screen-size}-{width}



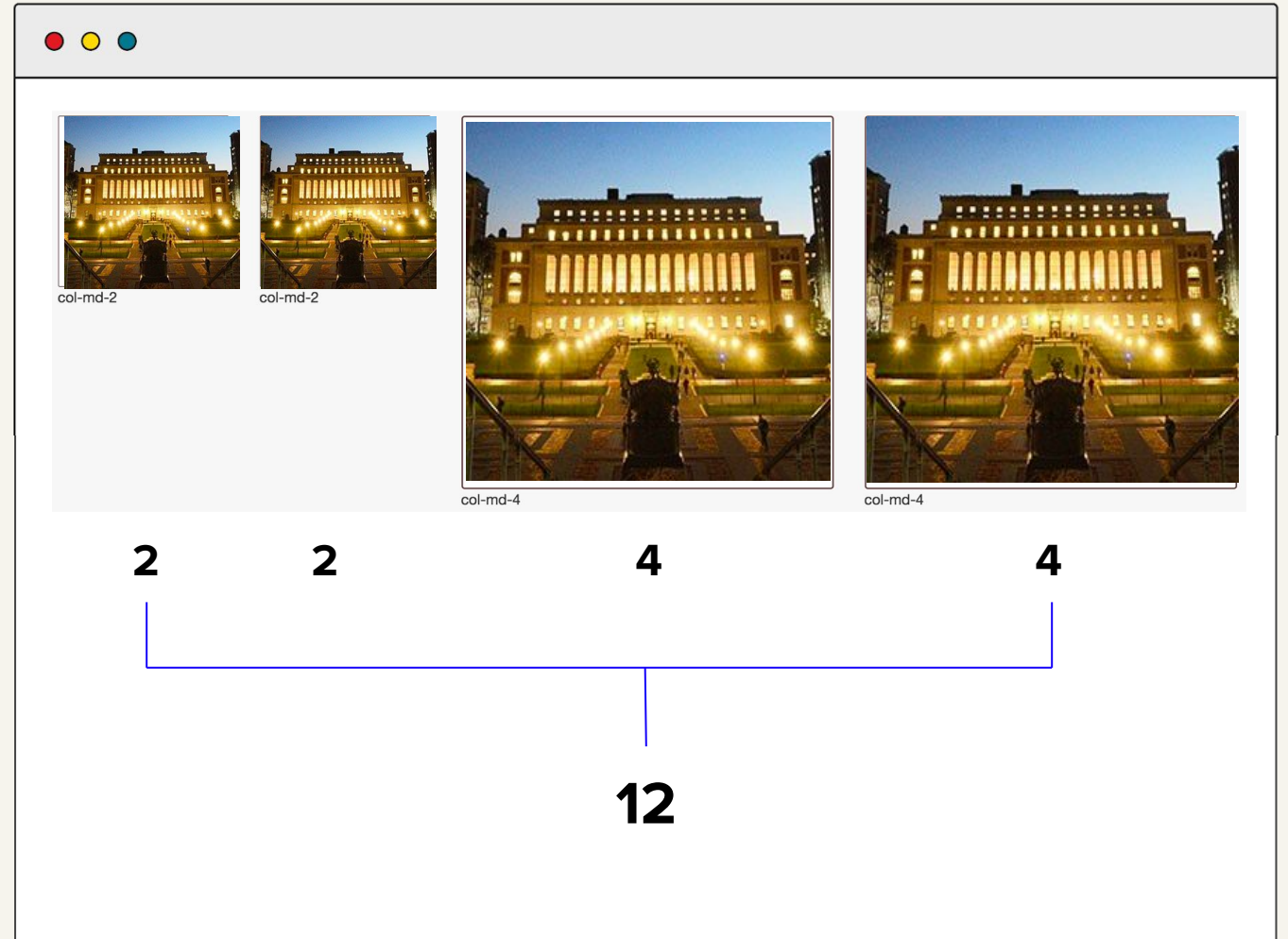
1. What will this produce?

```
<div class="container">
  <div class="row">
    <div class="col-md-3">
      
    </div>
    <div class="col-md-3">
      
    </div>
    <div class="col-md-3">
      
    </div>
    <div class="col-md-3">
      
    </div>
  </div>
</div>
```

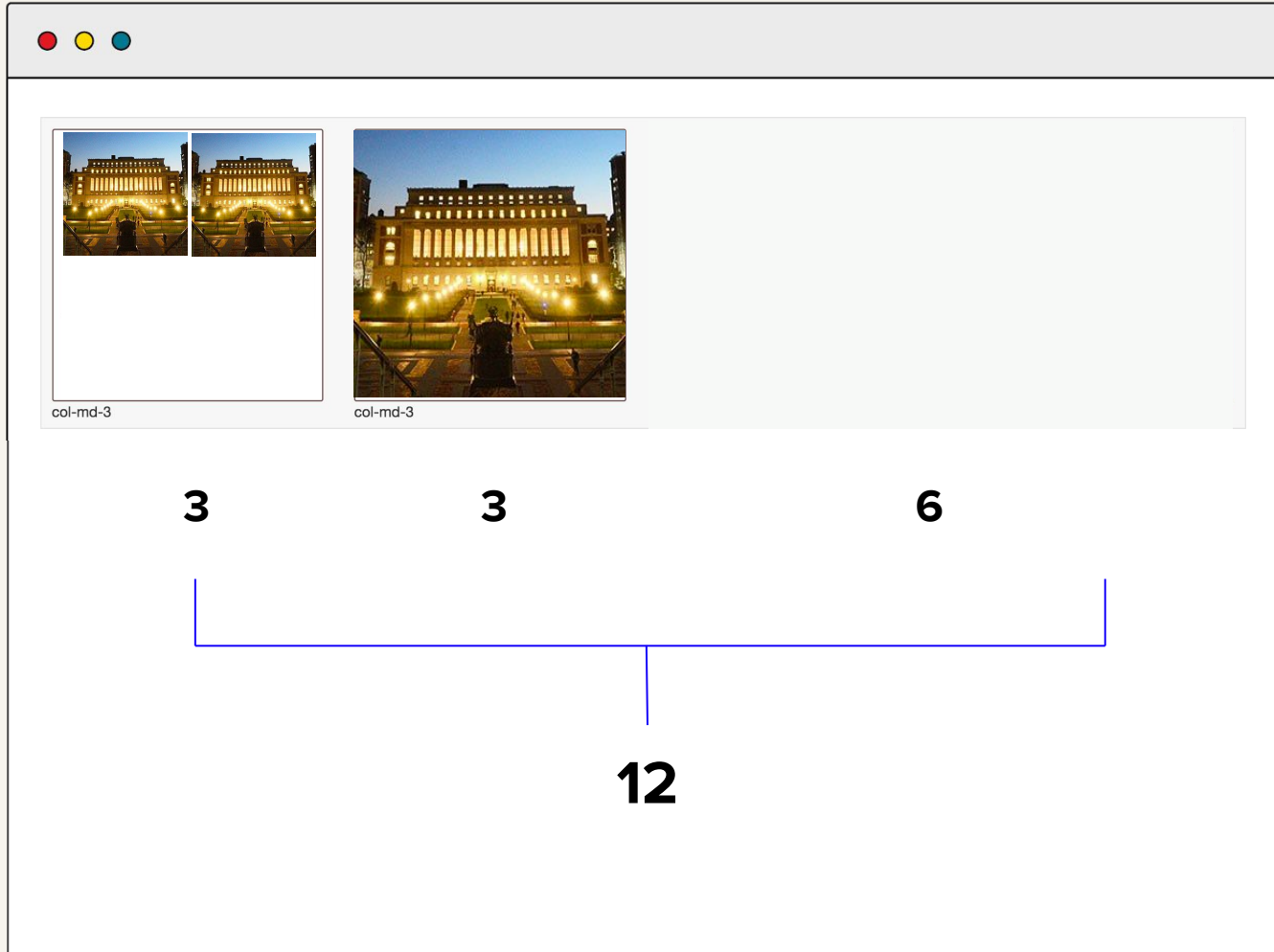


2. What will this produce?

```
<div class="container">
  <div class="row">
    <div class="col-md-2">
      
    </div>
    <div class="col-md-2">
      
    </div>
    <div class="col-md-4">
      
    </div>
    <div class="col-md-4">
      
    </div>
  </div>
</div>
```



3. What will this produce?



```
<div class="container">
  <div class="row">

    <div class="col-md-3">
      <div class="row">
        <div class="col-md-6">
          
        </div>
        <div class="col-md-6">
          
        </div>
      </div>
    </div>

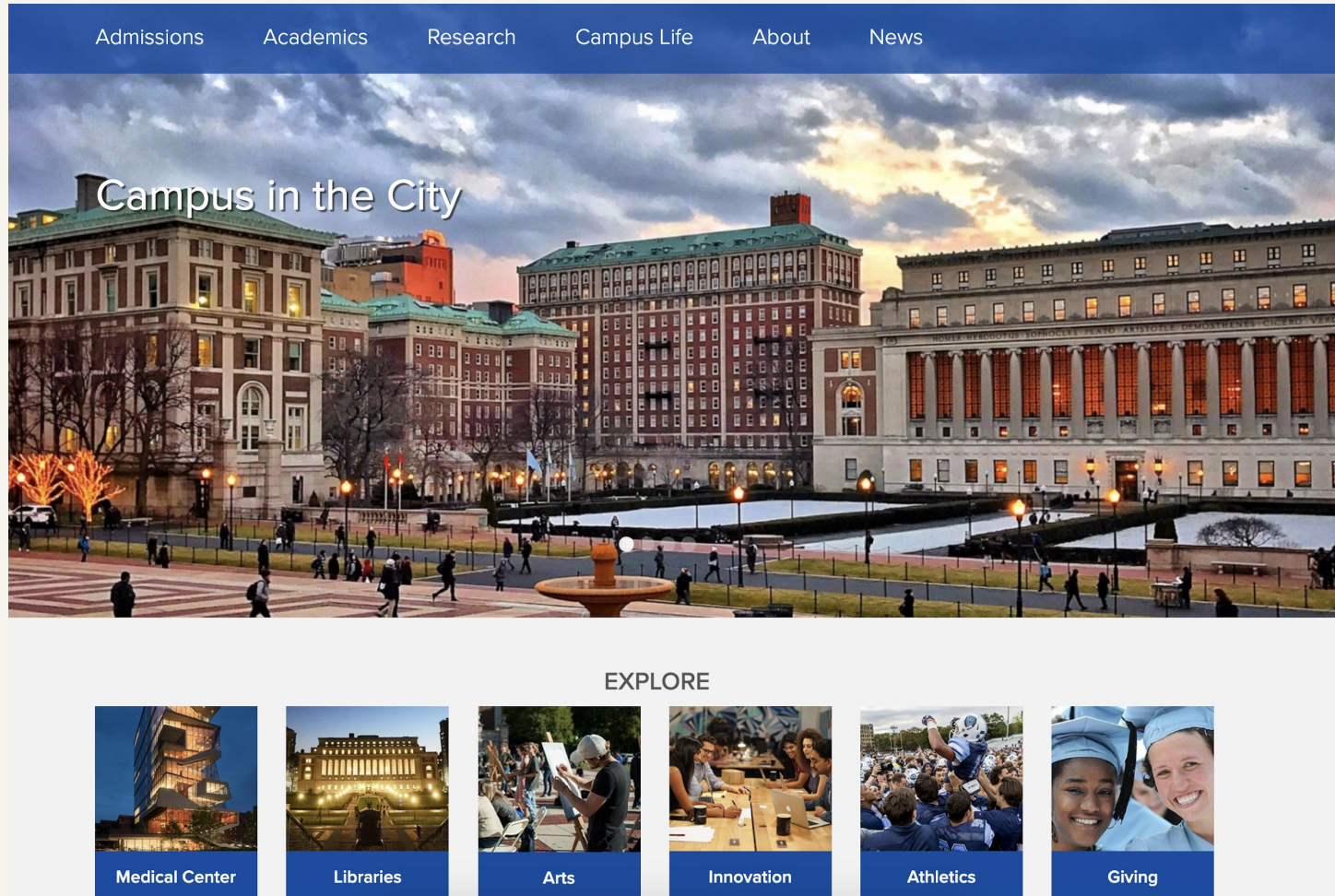
    <div class="col-md-3">
      
    </div>

    <div class="col-md-6"></div>

  </div>
</div>
```



Bootstrap solves another problem for you



Bootstrap enables responsive design, making it easy for the layout to respond to changes in the screen size.



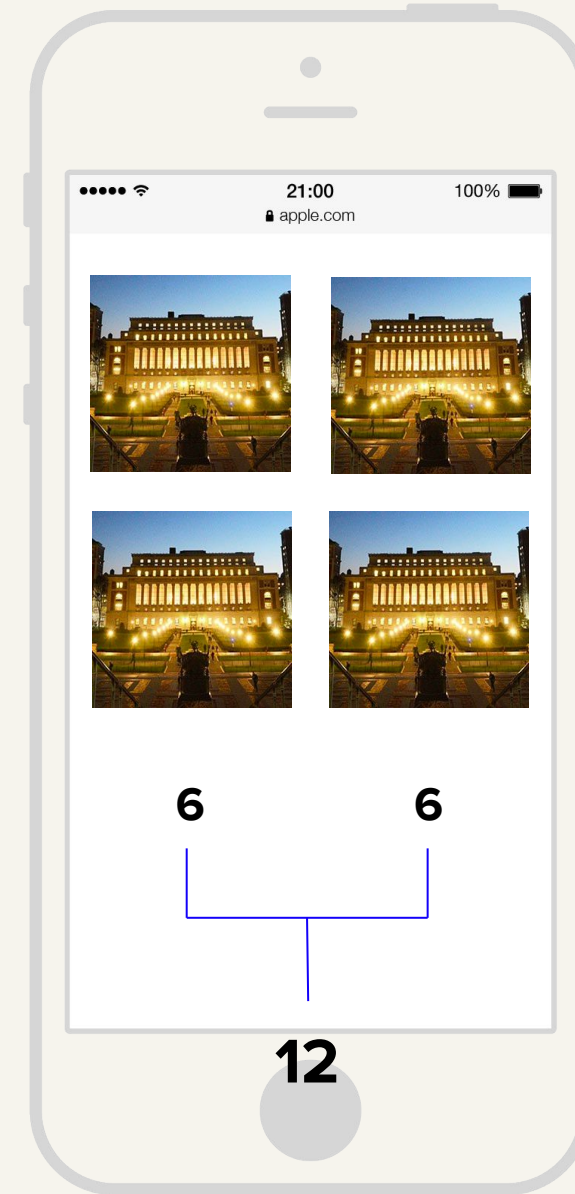
4 Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. **Follow platform conventions.**

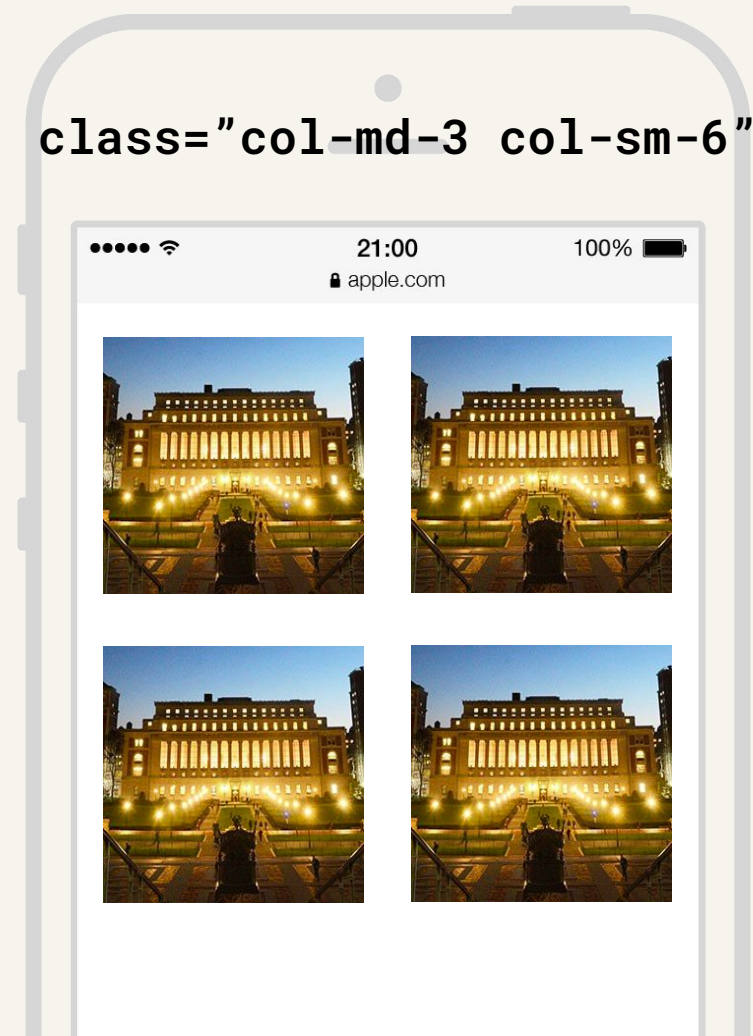


What will this do on a phone?

```
<div class="row">
  <div class="col-md-3 col-sm-6">
    
  </div>
  <div class="col-md-3 col-sm-6">
    <img src = "./columbia.jpg">
  </div>
  <div class="col-md-3 col-sm-6">
    <img src = "./columbia.jpg">
  </div>
  <div class="col-md-3 col-sm-6">
    <img src = "./columbia.jpg">
  </div>
</div>
```



What's the difference in layout?

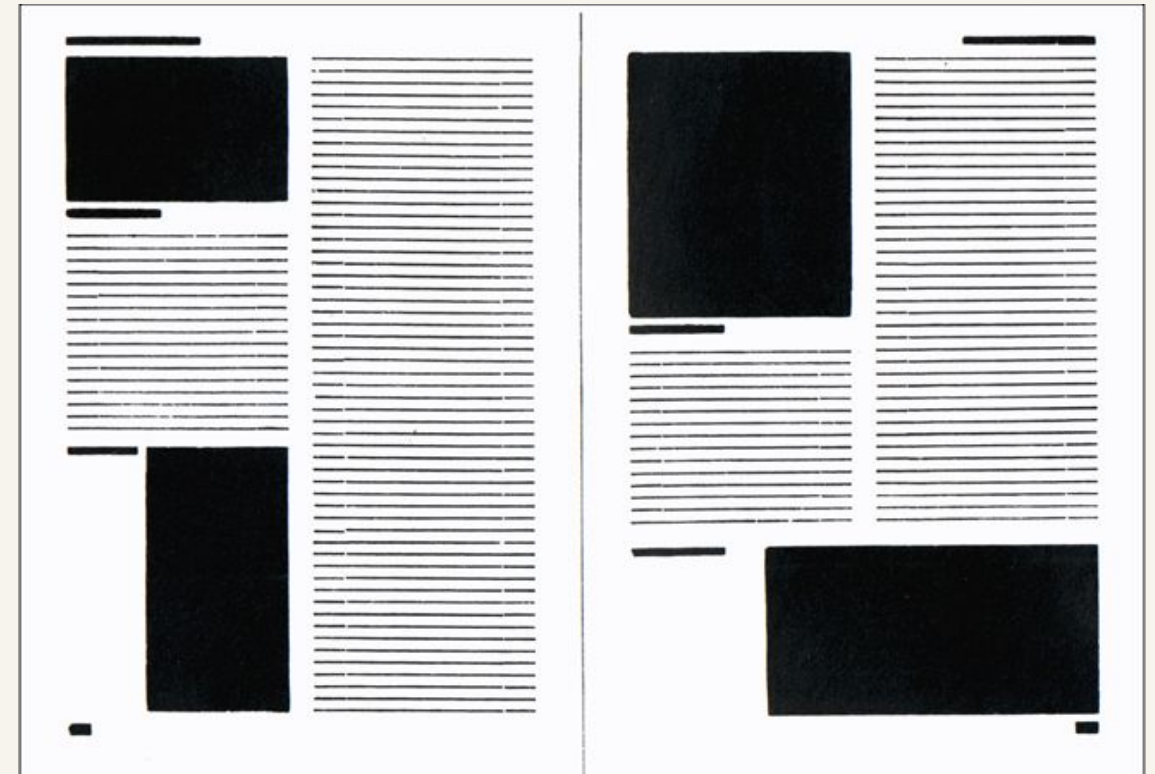
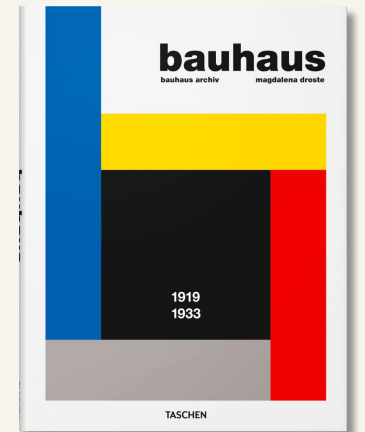


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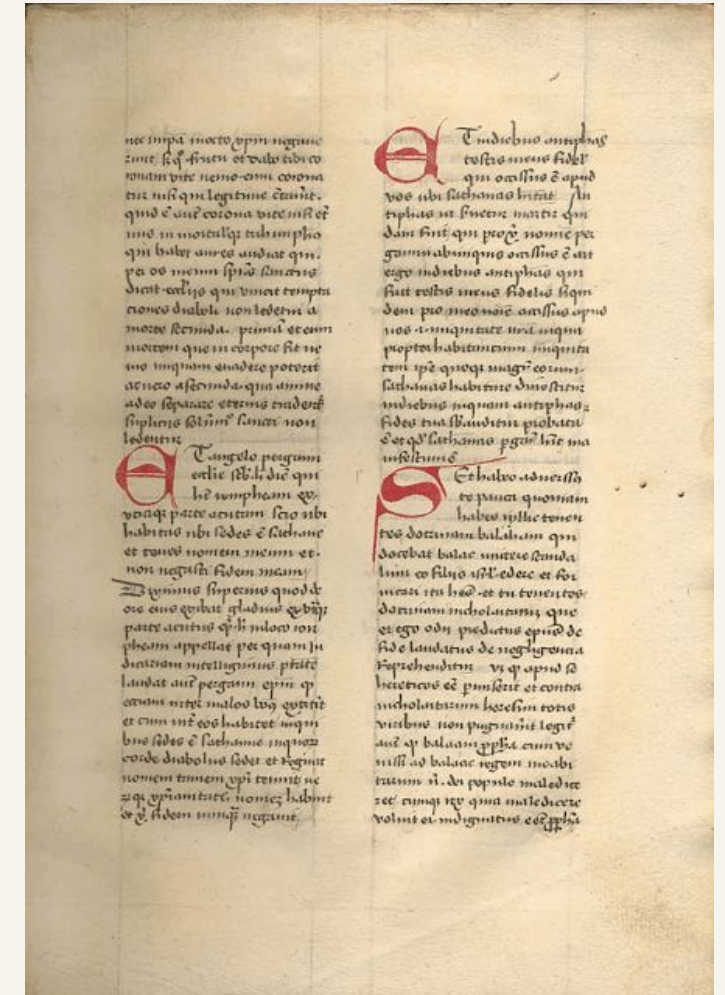
Summary



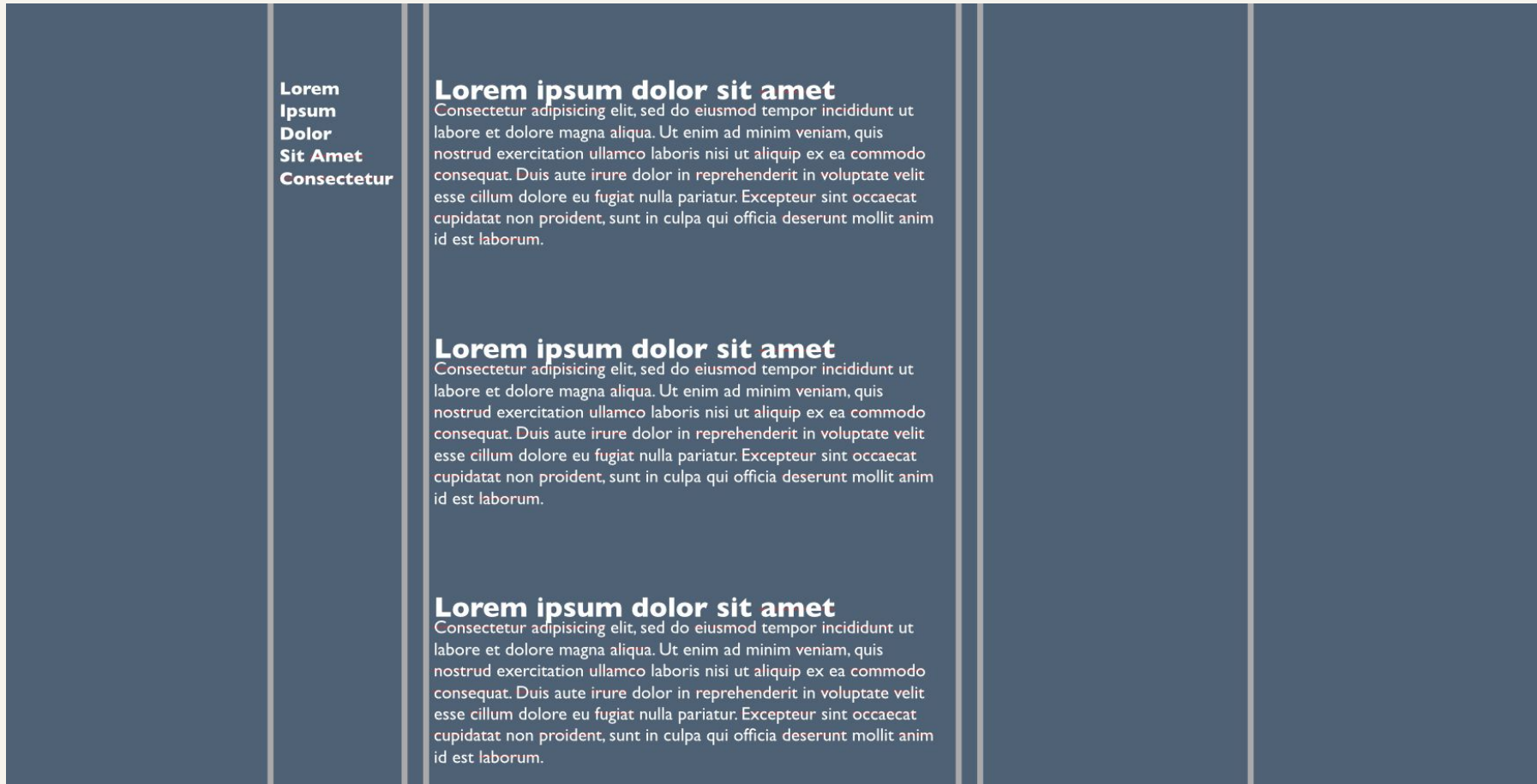
The Bauhaus School of Design helped give rise to asymmetric grid layouts.



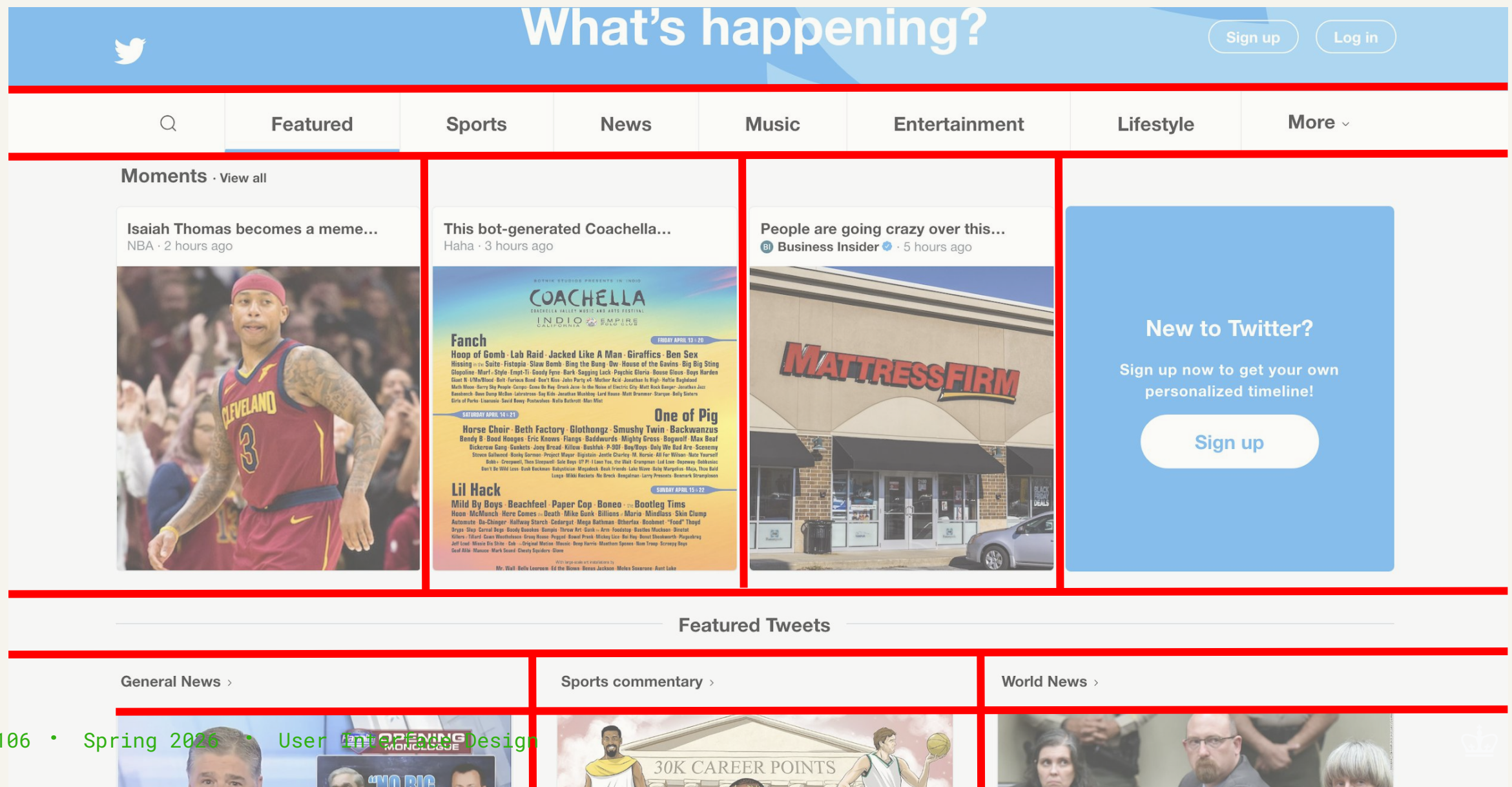
Historically, information layout was symmetrical



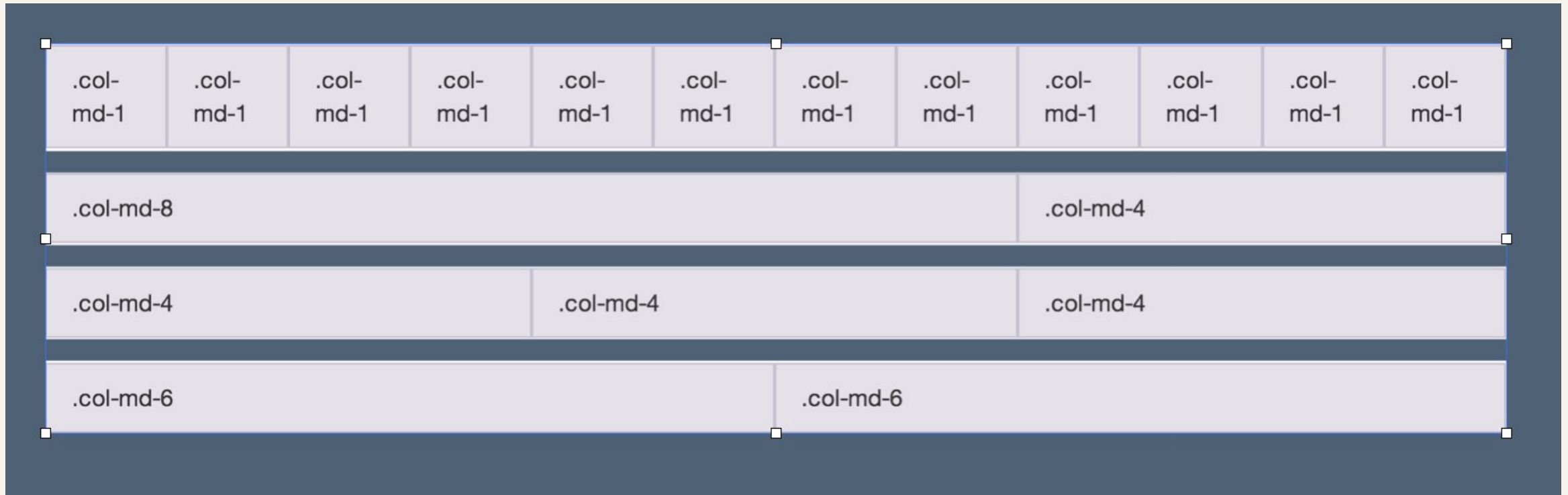
Grid layouts help you convey importance with grouping, location, size and whitespace.



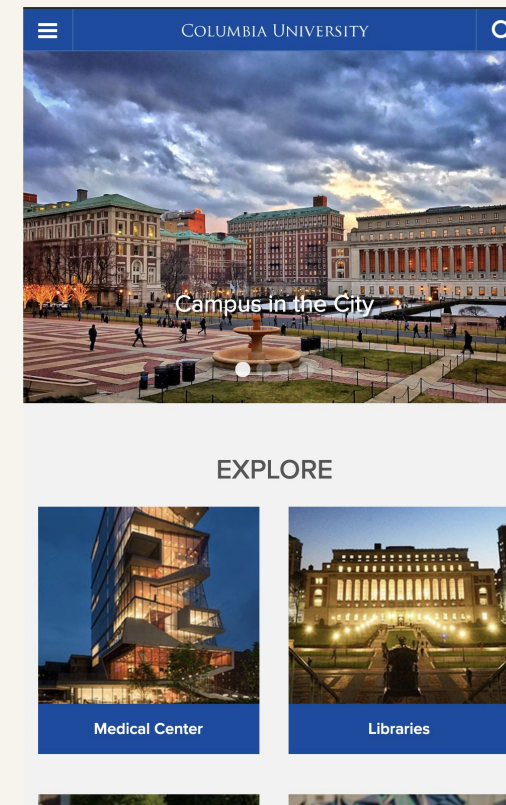
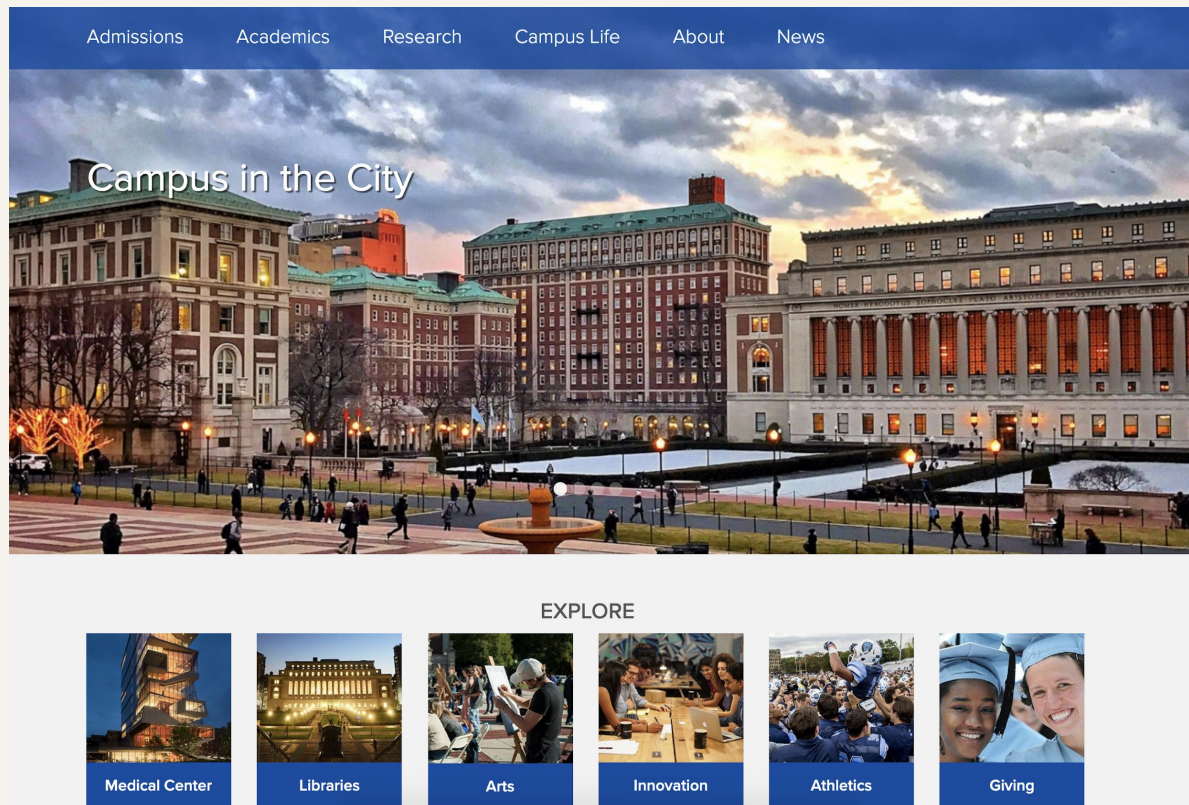
Twitter Bootstrap is a widely used grid system.



Bootstrap allocates width into twelve columns



Responsive design helps you specify the layout for different sized screens



Iterative Style of Programming helps you build a mental model of your code.

What is the **smallest** unit of progress I can make?

Iteration

```
<html>
  <body>
    Hello world!
  </body>
</html>
```

Goal

Hello World!



Does it look ok?



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Homework 02



Homework 02

Warm Up due **Friday at 11:59PM**

- You can complete this now

Main Assignment due **Monday at 11:59PM**

- You will need Tuesday's lecture

Homework is posted to Courseworks:
<https://courseworks2.columbia.edu/courses/237745>



